

Unit 7

EQUILIBRIA

56 items · 299 marks

Syllabus sections in this unit

7.1 Chemical equilibria: reversible reactions, dynamic equilibrium

7.2 Brønsted–Lowry theory of acids and bases

Unit 7: Equilibria**1. M/J 26/P21/Q2/b(i)**

2 The Contact process uses sulfur to produce sulfuric acid.

stage 1 Sulfur is burned in excess air.

stage 2 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$

stage 3 SO_3 reacts to produce H_2SO_4 .

(b) The reaction conditions for stage 2 include:

- using a ratio of 1 mol SO_2 to 1 mol O_2
- heat in the presence of V_2O_5 .

(i) Stage 2 is repeated using a ratio of more than 1 mol SO_2 to 1 mol O_2 .

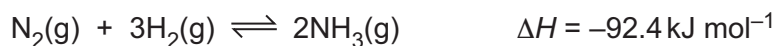
Predict the effect on the percentage yield of SO_3 at equilibrium. Use Le Chatelier's Principle to explain your answer.

[1]

2. M/J 26/P22/Q1/e(i)

1 Oxygen reacts with many elements.

(e) Ammonia is made in the Haber process.



A temperature of 400–450 °C is used in the Haber process.

(i) State **two** other conditions used in the Haber process.

1

2

[2]

3. M/J 26/P22/Q3/a

also Unit 11 (1 mark); this unit 2 marks

- 3 Separate samples of NaCl and NaBr are reacted with concentrated H₂SO₄. Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	- effervescence - steamy fumes that turn blue litmus paper red
2	NaBr	- effervescence - steamy fumes initially that turn blue litmus paper red - brown vapour - pungent smelling gas

- (a) (i) Write an equation for reaction 1.

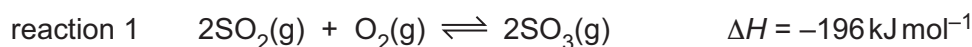
[1]

- (ii) Explain why reaction 1 is described as a Brønsted–Lowry acid–base reaction.

[2]

4. M/J 26/P23/Q3/b,c,d

- 3 The conditions used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process are carefully controlled.



- (b) Define Le Chatelier's principle.

[2]

- (c) In industry, reaction 1 occurs at 400–450 °C and 110–200 kPa with a catalyst.

Complete Table 3.1 by using the terms 'increase', 'decrease' or 'no change'.

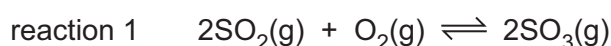
Table 3.1

	use temperature greater than 450 °C	without catalyst
effect on value of equilibrium constant, K_p		
effect on yield of product at equilibrium		
effect on rate of reaction		

[3]

- (d) A sealed tube that initially contains only 2.00 mol of SO₂(g) and 2.00 mol of O₂(g) is left to reach equilibrium.

At equilibrium, 1.96 mol of SO₃(g) is present and the total pressure is 150 kPa.



- (i) Complete Table 3.2.

Table 3.2

	SO ₂ (g)	O ₂ (g)	SO ₃ (g)
amount initially / mol	2.00	2.00	0
amount at equilibrium / mol			1.96
mole fraction at equilibrium	0.01325	0.3377	

[3]

- (ii) Table 3.3 shows the mole fractions at equilibrium of SO
- ₂
- (g), O
- ₂
- (g) and SO
- ₃
- (g) under different conditions.

Table 3.3

	SO ₂ (g)	O ₂ (g)	SO ₃ (g)
mole fraction at equilibrium	0.360	0.019	0.621

Calculate K_p at 200 kPa using the data from Table 3.3.

Include units in your answer.

$$K_p =$$

units

[3]

5. M/J 26/P24/Q2/b(ii),c

- (b) Nitrogen is used to manufacture ammonia, NH
- ₃
- , in the Haber process. An iron catalyst is used.

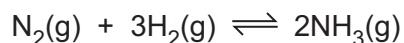
- (ii) State the temperature and pressure used in the Haber process.

temperature =

pressure =

[2]

- (c) The equation for the Haber process is shown.



Three separate experiments are carried out using mixtures of N₂(g) and H₂(g) to find the amount of NH₃(g) produced at equilibria under different conditions.

- (i) In experiment 1, 3.55 mol of N
- ₂
- (g) and 8.60 mol of H
- ₂
- (g) are mixed and left to reach equilibrium.

At equilibrium, 3.24 mol of NH₃ is present.

Complete Table 2.1.

Table 2.1

	N₂(g)	H₂(g)	NH₃(g)
initial amount/mol	3.55	8.60	0
change in amount/mol			3.24
amount at equilibrium/mol			3.24

[3]

- (ii) In experiment 2, under different conditions N₂(g) and H₂(g) are mixed and left to reach equilibrium.

Table 2.2 shows the amount, in mol, at equilibrium of N₂(g), H₂(g) and NH₃(g).

Table 2.2

	N₂(g)	H₂(g)	NH₃(g)
amount at equilibrium/mol	0.874	1.340	0.347

Calculate the mole fraction of NH₃(g) in the equilibrium mixture in experiment 2.

$$\text{mole fraction of NH}_3(\text{g}) =$$

[2]

- (iii) The expression for the equilibrium constant, K_p , for the Haber process is shown.

$$K_p = \frac{p(\text{NH}_3)^2}{p(\text{N}_2) \times p(\text{H}_2)^3}$$

Describe what is meant by partial pressure.

[1]

- (iv) In experiment 3, the mixture at equilibrium contains the following partial pressures of N₂(g) and H₂(g).

$$p(\text{N}_2) = 1590 \text{ kPa}$$

$$p(\text{H}_2) = 1750 \text{ kPa}$$

For experiment 3, the equilibrium constant, K_p , is $8.49 \times 10^{-7} \text{ kPa}^{-2}$.

Calculate the partial pressure, in kPa, of NH₃(g) at equilibrium.

$$p(\text{NH}_3) =$$

kPa [3]

6. MAR 26/P22/Q4/a(i,ii,iii)

4 Cyanoethanoic acid, $\text{CH}_2(\text{CN})\text{COOH}$, is a weak organic acid.

(a) At room temperature, $\text{CH}_2(\text{CN})\text{COOH}$ undergoes an acid–base reaction with $\text{NaOH}(\text{aq})$.

(i) Define weak acid.

[1]

(ii) Sketch the pH titration curve on Figure 4.1 for the reaction of:

- 25.00 cm^3 of 0.10 mol dm^{-3} $\text{CH}_2(\text{CN})\text{COOH}(\text{aq})$ and
- 0.10 mol dm^{-3} $\text{NaOH}(\text{aq})$, added to excess.

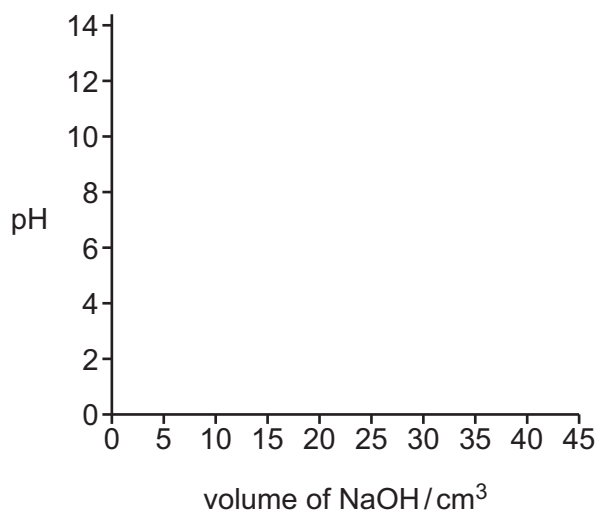


Figure 4.1

[3]

(iii) Write the overall ionic equation for the neutralisation of $\text{CH}_2(\text{CN})\text{COOH}(\text{aq})$ by $\text{NaOH}(\text{aq})$ in **4(a)(ii)**.

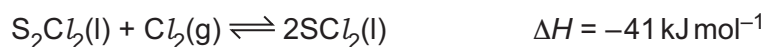
Include state symbols.

[1]

7. O/N 25/P22/Q2/e

2 The Period 3 elements show trends in physical and chemical properties across the period.

- (e) $\text{S}_2\text{Cl}_2(\text{l})$ reacts with $\text{Cl}_2(\text{g})$ in a reversible reaction to form $\text{SCl}_2(\text{l})$. Under certain conditions, a dynamic equilibrium is established.



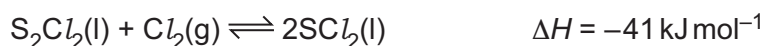
- (i) State what is meant by dynamic equilibrium.

[1]

- (ii) Identify the condition necessary to establish dynamic equilibrium.

[1]

- (iii) $\text{S}_2\text{Cl}_2(\text{l})$ is yellow and $\text{SCl}_2(\text{l})$ is red.



State what is observed when the following changes are made to an equilibrium mixture of $\text{S}_2\text{Cl}_2(\text{l})$ and $\text{SCl}_2(\text{l})$.

Explain your answers.

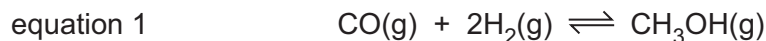
- The equilibrium mixture is warmed gently.
observation
explanation
- The overall pressure of the equilibrium mixture is increased.
observation
explanation

[4]

8. O/N 25/P24/Q2/b

2 Aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , are both used as reagents and catalysts.

(b) A 5.00 dm^3 sealed flask contains 0.400 mol of CO(g) and 0.800 mol of $\text{H}_2\text{(g)}$ and an Al_2O_3 catalyst. The flask is heated to a temperature of 290°C and allowed to reach equilibrium. Equation 1 shows the reaction.



The equilibrium constant, K_c , of equation 1 is given.

$$K_c = \frac{[\text{CH}_3\text{OH}]}{[\text{CO}][\text{H}_2]^2}$$

(i) State the units of K_c .

[1]

(ii) The equilibrium mixture contains 0.280 mol of $\text{CH}_3\text{OH(g)}$.

Calculate the value of K_c .

Give your answer to **three** significant figures.

value of $K_c =$ [3]

(iii) State and explain the effect, if any, on the value of K_c when the overall pressure in the sealed flask is increased.

[1]

9. M/J 25/P21/Q3/b,c

(b) Define Le Chatelier's principle.

[2]

(c) Reaction 2 shows the equilibrium reaction between X(g) and Y(g) to produce Z(g) in a sealed container.

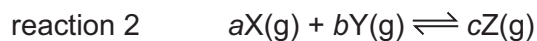


Fig. 3.3 shows the effect of changing pressure on the percentage yield of Z(g) at two different temperatures, 300 K and 350 K.

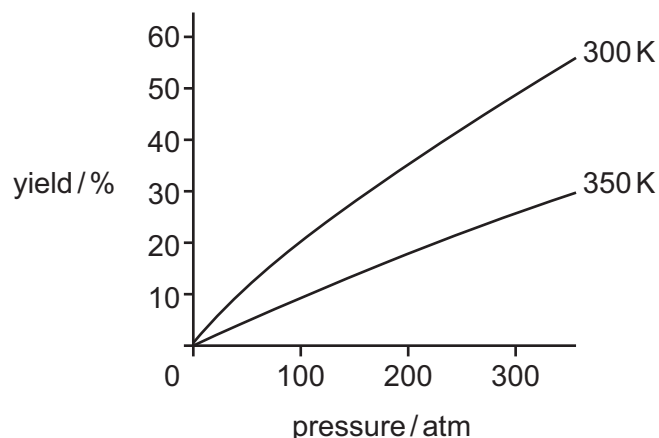


Fig. 3.3

Deduce **two** conclusions about reaction 2 using Fig 3.3.

[2]

10. M/J 25/P23/Q2/d

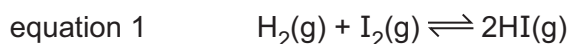
2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

(d) State why ZnO is described as a Brønsted–Lowry base when it is added to H₂SO₄(aq).

[1]

11. M/J 25/P24/Q3/b,c

- 3 $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ react to form $\text{HI}(\text{g})$ in a reversible reaction, as shown in equation 1.



In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

- (b) In experiment 2, 0.100 mol of $\text{H}_2(\text{g})$, 0.200 mol of $\text{I}_2(\text{g})$ and 0.300 mol of $\text{HI}(\text{g})$ are combined.

At equilibrium 0.154 mol of I_2 is present.

Calculate the amounts, in mol, of H_2 and HI in the equilibrium mixture produced in experiment 2.

amount of H_2 in equilibrium mixture = _____ mol

amount of HI in equilibrium mixture = _____ mol
[3]

- (c) In experiment 3, 0.0772 mol of H_2 and 0.0986 mol of I_2 are present in an equilibrium mixture at 298 K.



Use the equilibrium constant, K_c , to calculate the amount, in mol, of HI present in the equilibrium mixture in experiment 3.

amount of HI in equilibrium mixture = _____ mol [1]

12. O/N 24/P21/Q3/b(iii)

- 3 The Group 14 elements show a change from non-metallic to metallic character down the group.

- (b) Carbon is found in inorganic compounds such as carbonates.

- (iii) Ammonium carbonate undergoes an acid–base reaction with $\text{NaOH}(\text{aq})$.

Explain this statement.

[2]

13. O/N 24/P23/Q3/b(iii)

- 3 The Group 14 elements show a change from non-metallic to metallic character down the group.

- (b) Carbon is found in inorganic compounds such as carbonates.

- (iii) Ammonium carbonate undergoes an acid–base reaction with $\text{NaOH}(\text{aq})$.

Explain this statement.

[2]

14. M/J 24/P21/Q2/c

also Unit 6 (1 mark), Unit 10 (1 mark); this unit 2 marks

(c) Solid magnesium nitride, Mg_3N_2 , is a crystalline solid.

- (i) Deduce the oxidation numbers of magnesium and nitrogen in magnesium nitride to complete Table 2.1.

Table 2.1

	oxidation number in Mg_3N_2
magnesium	
nitrogen	

[1]

- (ii) Magnesium nitride reacts with an excess of water to produce ammonia and magnesium hydroxide only. Construct an equation to describe this reaction.

[1]

- (iii) Explain why the solution produced in the reaction in (c)(ii) has a pH greater than 7. Refer to the products of the reaction in your answer.

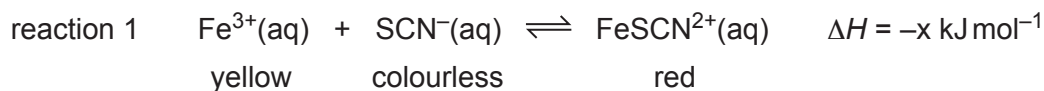
[2]

15. M/J 24/P21/Q3/a,b,c

3 (a) Define Le Chatelier's principle.

[2]

(b) Reaction 1 describes the reversible reaction between yellow $\text{Fe}^{3+}(\text{aq})$ and colourless $\text{SCN}^{-}(\text{aq})$ to produce red $\text{FeSCN}^{2+}(\text{aq})$.



A mixture of $\text{Fe}^{3+}(\text{aq})$, $\text{SCN}^{-}(\text{aq})$ and $\text{FeSCN}^{2+}(\text{aq})$ is at equilibrium at 20°C .

The temperature of this mixture is then increased to 50°C and allowed to reach equilibrium.

Deduce the changes that occur, if any, in the equilibrium mixture at 50°C compared to the equilibrium mixture at 20°C .

- change in appearance
- change in relative concentration of $\text{FeSCN}^{2+}(\text{aq})$
- change in value of the equilibrium constant, K_c

[3]

(c) In another experiment, equimolar amounts of $\text{Fe}^{3+}(\text{aq})$ and $\text{SCN}^{-}(\text{aq})$ are mixed together and allowed to reach equilibrium. The total volume of the mixture is 25.0 cm^3 .



At equilibrium the mixture contains:

- $[\text{SCN}^{-}] = 1.30 \times 10^{-3} \text{ mol dm}^{-3}$
- $[\text{FeSCN}^{2+}] = 0.300 \times 10^{-3} \text{ mol dm}^{-3}$.

(i) Calculate the initial amount, in mol, of $\text{Fe}^{3+}(\text{aq})$ added to $\text{SCN}^{-}(\text{aq})$ to produce this mixture.

initial amount of $\text{Fe}^{3+}(\text{aq}) =$ mol [2]

(ii) Calculate K_c for reaction 1 and state its units.

Show your working.

$K_c =$

units

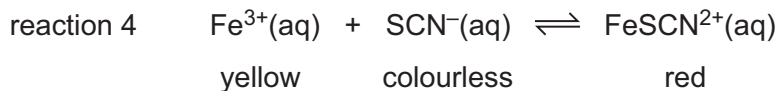
[2]

16. M/J 24/P22/Q3/a,b

- 3 (a)** Describe what is meant by dynamic equilibrium.

[2]

- (b)** Reaction 4 describes the reversible reaction between yellow $\text{Fe}^{3+}(\text{aq})$ and colourless $\text{SCN}^{-}(\text{aq})$ to produce red $\text{FeSCN}^{2+}(\text{aq})$.



An equilibrium mixture contains $\text{Fe}^{3+}(\text{aq})$, $\text{SCN}^{-}(\text{aq})$ and $\text{FeSCN}^{2+}(\text{aq})$. A few colourless crystals of soluble $\text{KSCN}(\text{s})$ are added. The mixture is then left until it reaches equilibrium again. The temperature of both equilibrium mixtures is the same.

- (i)** Deduce the changes that occur, if any, in the equilibrium mixture after $\text{KSCN}(\text{s})$ is added compared to the original equilibrium mixture.
- change in appearance
 - change in relative concentration of $\text{Fe}^{3+}(\text{aq})$
 - change in value of the equilibrium constant, K_c

[3]



- (ii)** The expression for the equilibrium constant, K_c , for reaction 4 is shown.

$$K_c = \frac{[\text{FeSCN}^{2+}(\text{aq})]}{[\text{Fe}^{3+}(\text{aq})] \times [\text{SCN}^{-}(\text{aq})]}$$

$5.00 \times 10^{-5} \text{ mol}$ of $\text{Fe}^{3+}(\text{aq})$ and $5.00 \times 10^{-5} \text{ mol}$ of $\text{SCN}^{-}(\text{aq})$ are added together and allowed to reach equilibrium. The total volume of the mixture is 25.0 cm^3 .

At equilibrium the concentration of $\text{FeSCN}^{2+}(\text{aq})$ is $4.23 \times 10^{-4} \text{ mol dm}^{-3}$.

Calculate the equilibrium constant, K_c , for reaction 4.

Include the units in your answer.

 $K_c =$

units

[4]



17. M/J 24/P23/Q2/a,b(i),c

2 $\text{N}_2(\text{g})$ reacts with $\text{H}_2(\text{g})$ in the Haber process, as shown in reaction 1.

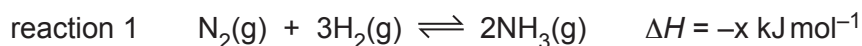


Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	B	C
initial molar ratio of $\text{N}_2 : \text{H}_2$ added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $\text{NH}_3(\text{g})$ at equilibrium	58	x	y

(a) Describe and explain the change, if any, to the percentage yield of $\text{NH}_3(\text{g})$ produced in **B** compared to **A**.

[1]

(b) (i) Describe and explain the change, if any, to the percentage yield of $\text{NH}_3(\text{g})$ produced in **C** compared to **A**.

[1]

(c) (i) Write an expression for the equilibrium constant, K_p , for reaction 1. State the units.

$$K_p =$$

units

[2]

(ii) Equilibrium mixture **D** is made when 1.0 mol of $\text{N}_2(\text{g})$ and 3.0 mol of $\text{H}_2(\text{g})$ are added to a sealed container at 750 °C and 1000 atm and left to reach equilibrium. This mixture contains 1.16 mol of $\text{NH}_3(\text{g})$.

Calculate the mole fraction of $\text{NH}_3(\text{g})$ in **D**.

$$\text{mole fraction of } \text{NH}_3(\text{g}) =$$

[2]

(iii) The mole fraction of $\text{N}_2(\text{g})$ is 0.625 in a new equilibrium mixture, **E**.

Calculate the partial pressure of $\text{N}_2(\text{g})$ in **E** when the total pressure is 1000 atm.

$$\text{partial pressure of } \text{N}_2(\text{g}) =$$

atm [1]

18. M/J 24/P23/Q4/b(i)

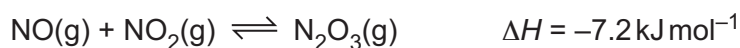
(b) An exothermic reaction occurs when $\text{NH}_4^+(\text{aq})$ is added to $\text{OH}^-(\text{aq})$.

(i) Identify the type of reaction.

[1]

19. O/N 23/P22/Q2/a,b,c

2 NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

(a) Fig. 2.1 shows how the rate of the forward reaction changes with time.

Initially, the rate of the reverse reaction is zero.

Complete Fig. 2.1 to sketch how the rate of the **reverse** reaction changes with time.

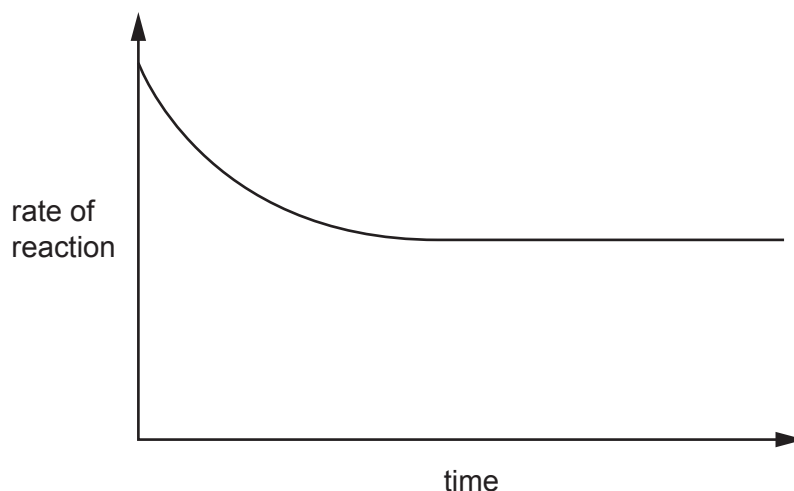


Fig. 2.1

[1]

(b) State how the position of equilibrium changes, if at all, when the reaction takes place at 100 °C.

Explain your answer.

Assume the pressure remains constant.

[2]

(c) Table 2.1 shows the composition of an equilibrium mixture of NO(g), NO₂(g) and N₂O₃(g) at 101 kPa.

Table 2.1

gas	number of moles at equilibrium/mol
NO	0.605
NO ₂	0.605
N ₂ O ₃	0.390

Calculate K_p , the equilibrium constant with respect to partial pressures.

Deduce the units of K_p .

$$K_p =$$

units

[3]

20. O/N 23/P22/Q3/a(ii)

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

(a) H_3PO_4 is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form H_3PO_4 .

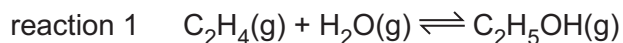
(ii) H_3PO_4 is a weak Brønsted–Lowry acid.

Define weak Brønsted–Lowry acid.

[2]

21. M/J 23/P22/Q4/b,c(i)

- 4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



- (b) Reaction 1 reaches equilibrium at constant temperature and pressure.

Deduce what effect increasing the pressure will have on the amount of ethanol in the new equilibrium mixture. Use Le Chatelier's principle to explain your answer.

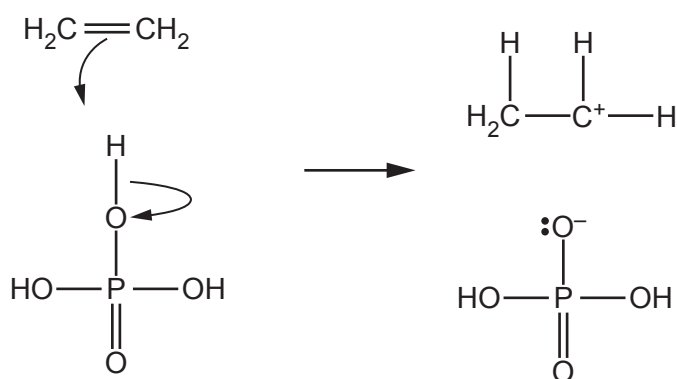
effect of increasing pressure

explanation

[2]

- (c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



step 2

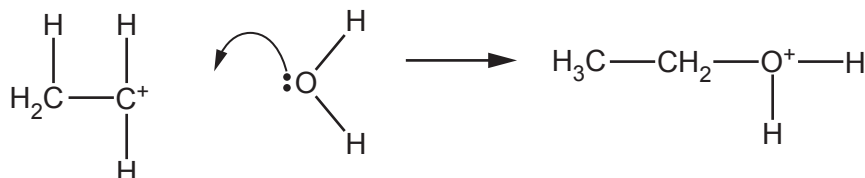


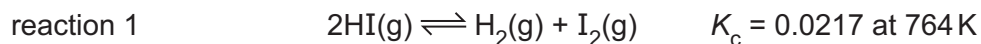
Fig. 4.1

- (i) Describe the behaviour of H_3PO_4 in step 1 in Fig. 4.1. Explain your answer.

[1]

22. M/J 23/P23/Q4/b,c

- (b) A sample of HI(g) is added to a 2.00 dm³ sealed vessel at 764 K and allowed to reach equilibrium.



At equilibrium the mixture contains 1.70 mol of HI(g).

- (i) State **one** difference in the appearance of the initial reaction mixture compared to the mixture at equilibrium.

[1]

- (ii) Deduce the expression for equilibrium constant K_c for reaction 1.

$$K_c =$$

[1]

- (iii) Calculate the concentration of I₂ present in the reaction mixture at equilibrium. Show your working.

$$\text{concentration of I}_2 = \quad \text{mol dm}^{-3} \quad [3]$$

- (c) The experiment is repeated at 500 K. The value of K_c under these conditions is 0.00625.

- (i) Describe the difference in the composition of the equilibrium mixture at 500 K compared to 764 K.

[1]

- (ii) Use Le Chatelier's principle to deduce whether the decomposition of HI(g) is endothermic or exothermic. Explain your answer.

[1]

23. MAR 23/P22/Q3/d(ii)

- 3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

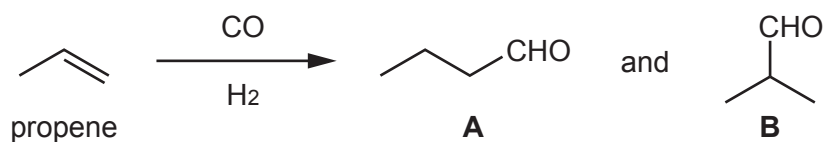


Fig. 3.1

- (d) The reaction of ethene, C₂H₄, with a 1:1 mixture of CO and H₂ is shown in equation 1.



At atmospheric pressure a cobalt-based catalyst is used in this reaction.

- (ii) Explain why the yield of CH₃CH₂CHO(g) increases when the overall pressure of the reaction mixture is increased.

[1]

24. O/N 22/P22/Q1/c(i)

1 Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

(c) NH_4^+ is a Brønsted–Lowry acid.

(i) Define Brønsted–Lowry acid.

[1]

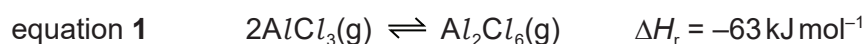
25. O/N 22/P22/Q2/c(ii)

2 The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl_3	SiCl_4	PCl_5	PCl_3	SCl_2
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

(c) In the gas phase, $\text{AlCl}_3(\text{g})$ exists at equilibrium with $\text{Al}_2\text{Cl}_6(\text{g})$ as shown.



(ii) State the effect of an increase in temperature on the equilibrium mixture in equation 1. Explain your answer.

[1]

26. M/J 22/P21/Q2/d

2 Nitrogen molecules, $\text{N}_2(\text{g})$, contain two atoms attracted to each other by a triple covalent bond.

(d) 25 cm^3 of 0.10 mol dm^{-3} $\text{HCl}(\text{aq})$ is added to a beaker and its pH is recorded.

50 cm^3 of 0.10 mol dm^{-3} $\text{NH}_3(\text{aq})$ is added to the $\text{HCl}(\text{aq})$ in 5 cm^3 portions.

The pH of the mixture is monitored until all the $\text{NH}_3(\text{aq})$ is added.

HCl is a strong Brønsted-Lowry acid.

(i) Describe what is meant by a strong Brønsted-Lowry acid.

[2]

(ii) NH_3 is a weak base.

Construct an equation that shows the behaviour of NH_3 as a weak Brønsted-Lowry base when dissolved in water.

[1]

(iii) On Fig. 2.1 sketch a graph to show the change in pH which occurs when $\text{HCl}(\text{aq})$ is titrated with $\text{NH}_3(\text{aq})$ as described in **(d)**.

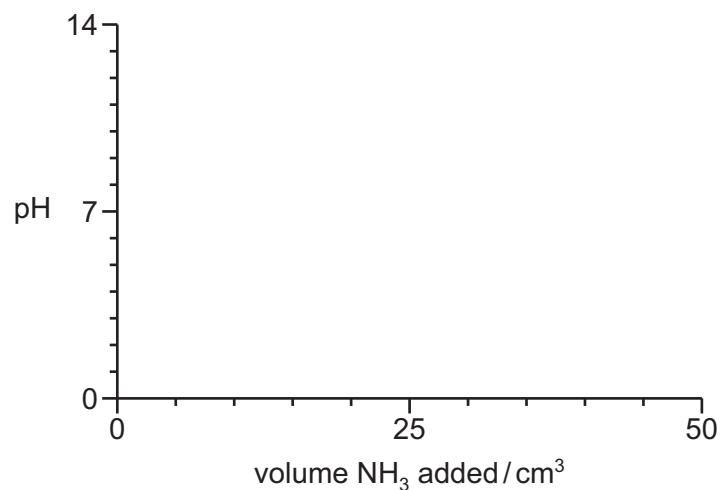
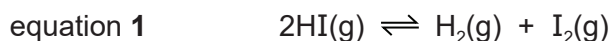


Fig. 2.1

[2]

27. M/J 22/P22/Q3/a,c

- 3 (a)** 0.025 mol of HI(g) is added to a closed vessel and left to reach dynamic equilibrium. The total pressure of the vessel is 100 kPa.



- (i)** Explain what is meant by dynamic equilibrium.

[2]

- (ii)** Describe **one** difference in the initial appearance of the reaction mixture compared to the mixture at equilibrium.

[1]

- (iii)** Write an expression for K_p for the reaction described in equation 1.

$$K_p =$$

[1]

- (iv)** At equilibrium the partial pressure of HI(g) is 86.4 kPa.

Calculate the amount of HI(g) present in the mixture at equilibrium. Show your working.

amount of HI(g) = mol [2]

- (c)** Describe the effect of increasing pressure on the value of K_p for the decomposition of HI(g).

[1]

28. M/J 22/P23/Q3/a(iii)

3 **G** belongs to a group of compounds called ethers.

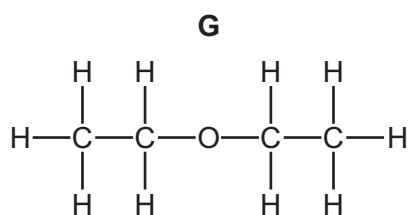
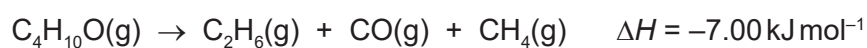
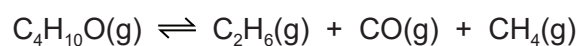


Fig. 3.1

(a) When **G**, $\text{C}_4\text{H}_{10}\text{O}$, is heated, thermal decomposition occurs.



(iii) When **G**, $\text{C}_4\text{H}_{10}\text{O}$, is heated in a sealed container, an equilibrium mixture is produced.



Complete the expression for the equilibrium constant, K_c , for this reaction.
State the units of K_c .

$K_c =$

units =

[2]

29. MAR 22/P22/Q2/d(iii)

2 Some oxides of elements in Period 3 are shown.



(d) SO_2 and SO_3 are found in the atmosphere.

The oxidation of SO_2 to SO_3 in the atmosphere is catalysed by NO_2 .

The first step of the catalytic oxidation is shown in equation 1.



(iii) Fig. 2.2 shows how the temperature of the atmosphere varies with height from the ground.

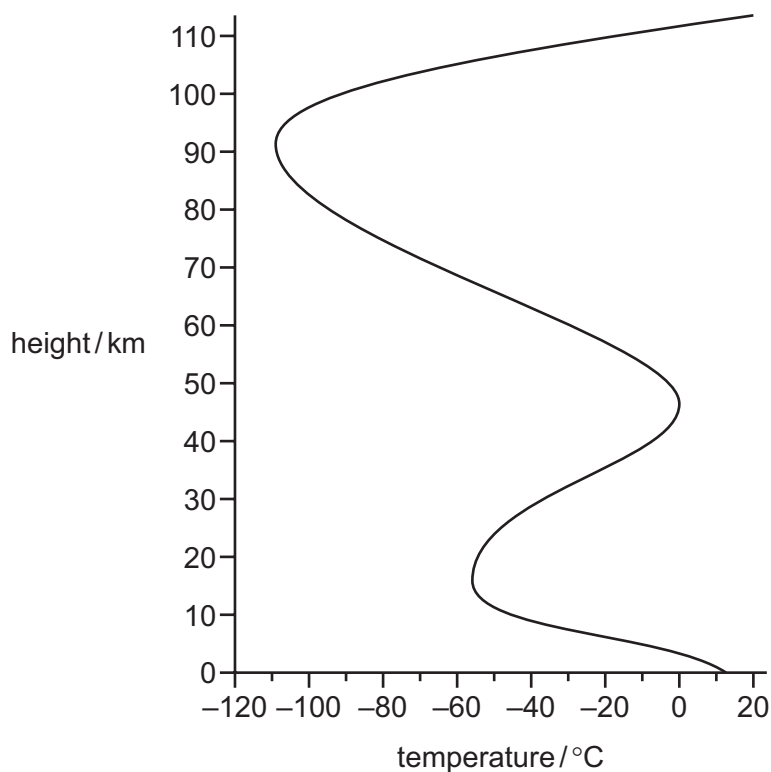


Fig. 2.2

The equilibrium reaction in equation 1 has $\Delta H^\circ = -168 \text{ kJ mol}^{-1}$.

Suggest how the position of this equilibrium differs at a height of 20 km compared with a height of 50 km from the ground.

Explain your answer.

[2]

30. MAR 22/P22/Q3/d

3 The hydrogen halides HCl, HBr and HI are all colourless gases at room temperature.

(d) The hydrogen halides dissolve in water to form strong Brønsted–Lowry acids.

The concentration of a strong acid can be determined by titration.

(i) State what is meant by strong Brønsted–Lowry acid.

[2]

(ii) On Fig. 3.2, sketch the pH titration curves produced when:

- 0.1 mol dm^{-3} NaOH(aq) is added to 25 cm^3 of 0.1 mol dm^{-3} HBr(aq), to excess
- 0.1 mol dm^{-3} NH_3 (aq) is added to 25 cm^3 of 0.1 mol dm^{-3} HBr(aq), to excess.

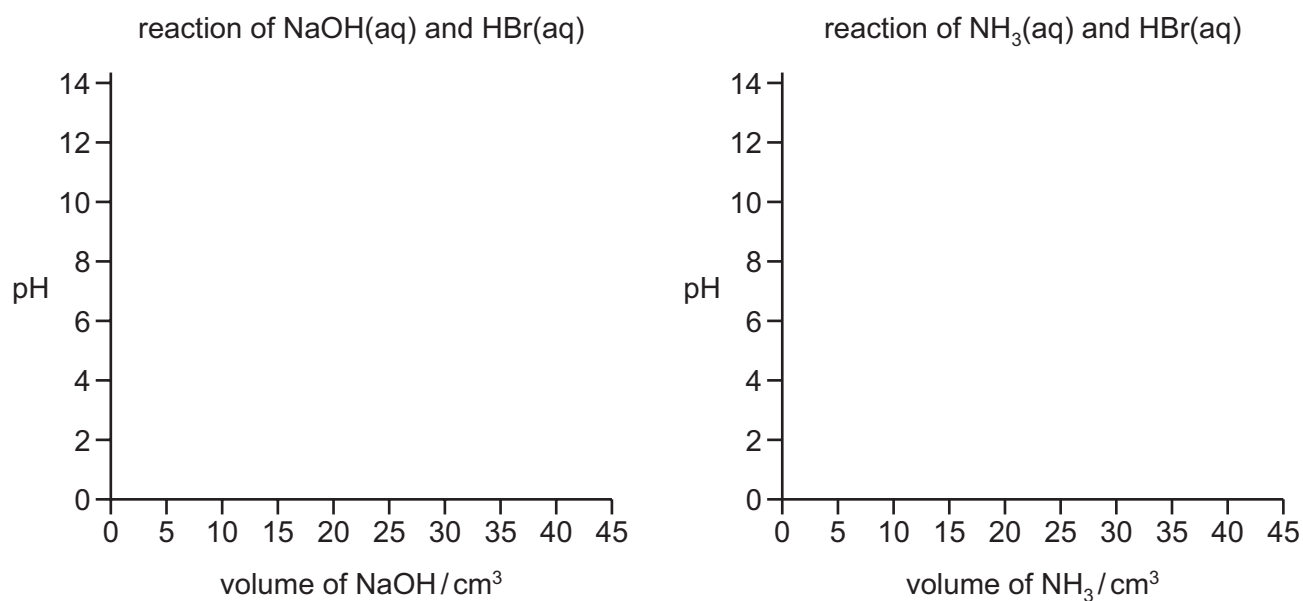


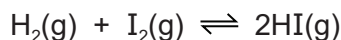
Fig. 3.2

[3]

31. O/N 21/P22/Q1/c,f(iii,iv)

1 Hydrogen iodide, HI, is a colourless gas at room temperature.

- (c)** HI(g) can be formed by reacting H₂(g) with I₂(g). The reaction is reversible, and an equilibrium forms quickly at high temperatures.



- (i)** Construct an expression for the equilibrium constant, K_p , for the reaction of H₂(g) and I₂(g) to form HI(g).

$$K_p =$$

[1]

- (ii)** The equilibrium partial pressures of the gases at 200 °C are as follows.

$$p_{\text{H}_2(\text{g})} = 895 \text{ Pa}$$

$$p_{\text{I}_2(\text{g})} = 895 \text{ Pa}$$

$$p_{\text{HI}(\text{g})} = 4800 \text{ Pa}$$

Calculate K_p for this reaction.

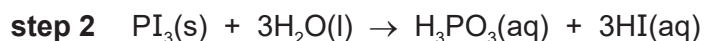
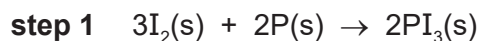
$$K_p =$$
 [1]

- (iii)** State how the value of K_p would change, if at all, if the reaction were carried out at 100 °C rather than 200 °C.

Explain your answer.

[2]

- (f)** In the laboratory, HI(aq) can be formed in a two-step process.



- (iii)** H₃PO₃(aq) and HI(aq) are both strong Brønsted–Lowry acids.

Give the meaning of the term *strong Brønsted–Lowry acid*

[2]

- (iv)** Give the formula of the conjugate base of H₃PO₃.

[1]

32. M/J 21/P23/Q1/d

1 The elements in Group 17 are known as the halogens.

(d) When $\text{ClO}^-(\text{aq})$ is added to water, it behaves as a Brønsted-Lowry base.

(i) Define the term *Brønsted-Lowry base*

[1]

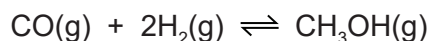
(ii) Write an ionic equation for the reaction between ClO^- and H_2O .

[1]

33. M/J 21/P23/Q2/c

2 Methanol, CH_3OH , is soluble in water because it forms hydrogen bonds with water molecules.

(c) Methanol is made by reacting carbon monoxide with hydrogen.



Carbon monoxide and hydrogen react at $1.0 \times 10^7 \text{ Pa}$ and 200°C . Eventually the reaction mixture reaches dynamic equilibrium.

The table shows the amounts of each species present in the mixture.

	$\text{CO}(\text{g})$	$\text{H}_2(\text{g})$	$\text{CH}_3\text{OH}(\text{g})$
initial amount/mol	1.0	2.0	0
equilibrium amount/mol	0.030	0.060	0.97

(i) Explain what is meant by *dynamic equilibrium*.

[2]

(ii) Calculate the partial pressure of methanol vapour at equilibrium under these conditions. Show your working.

Pa [2]

(iii) Write an expression for the equilibrium constant, K_p , for this reaction. State the units in your answer.

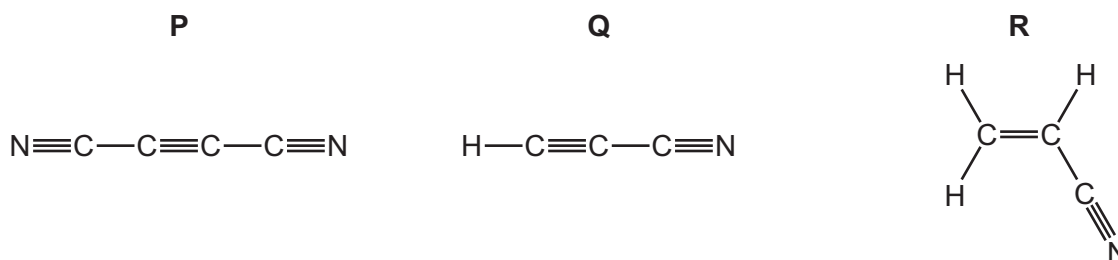
$K_p =$

units =

[2]

34. MAR 21/P22/Q3/b(i)

3 Compounds **P**, **Q** and **R** have all been found in the atmosphere of one of Saturn's moons.



(b) **Q** forms when HCN reacts with ethyne, $\text{H}-\text{C}\equiv\text{C}-\text{H}$.

(i) Ethyne, HCN and **Q** are all weak Brønsted–Lowry acids.

Explain what is meant by the term *weak Brønsted–Lowry acid*.

[2]

35. O/N 20/P21/Q2/b(i),d(iii)

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

(b) The two most common oxides of sulfur are SO_2 and SO_3 .

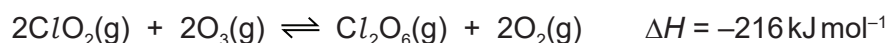
When SO_2 dissolves in water, a small proportion of it reacts with water to form a weak Brønsted–Lowry acid.

(i) Explain the meaning of the term *weak Brønsted–Lowry acid*.

[2]

(d) Chlorine forms several oxides, including Cl_2O , ClO_2 and Cl_2O_6 .

(iii) $\text{Cl}_2\text{O}_6(\text{g})$ is produced by the reaction of $\text{ClO}_2(\text{g})$ with $\text{O}_3(\text{g})$.



The reaction takes place at 500K and 100 kPa.

State and explain the effect on the yield of $\text{Cl}_2\text{O}_6(\text{g})$ when the experiment is carried out:

- at 1000K and 100 kPa
- at 500K and 500 kPa.

[4]

36. O/N 20/P22/Q3/d(ii)

3 PCl_5 , PCl_3 and NCl_3 are halides of Group 15 elements.

(d) NCl_3 is a yellow liquid that can be used to bleach flour.

(ii) NCl_3 reacts with water to form HOCl , a weak Brønsted–Lowry acid.

Explain fully what is meant by the term *weak Brønsted–Lowry acid*.

[2]

37. O/N 20/P23/Q2/b(i),d(iii)

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

(b) The two most common oxides of sulfur are SO_2 and SO_3 .

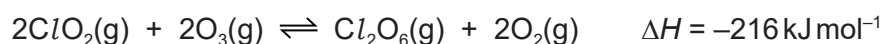
When SO_2 dissolves in water, a small proportion of it reacts with water to form a weak Brønsted-Lowry acid.

(i) Explain the meaning of the term *weak Brønsted-Lowry acid*.

[2]

(d) Chlorine forms several oxides, including Cl_2O , ClO_2 and Cl_2O_6 .

(iii) $\text{Cl}_2\text{O}_6(\text{g})$ is produced by the reaction of $\text{ClO}_2(\text{g})$ with $\text{O}_3(\text{g})$.



The reaction takes place at 500K and 100 kPa.

State and explain the effect on the yield of $\text{Cl}_2\text{O}_6(\text{g})$ when the experiment is carried out:

- at 1000K and 100 kPa
- at 500K and 500 kPa.

[4]

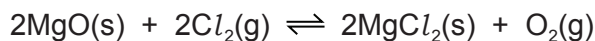
38. MAR 20/P22/Q1/a(ii),f

1 Group 2 metals form alkaline solutions in water.

(ii) Identify the ion that causes an aqueous solution to be alkaline.

[1]

(f) Magnesium oxide reacts reversibly with chlorine according to the following equation.



Under certain conditions, a dynamic equilibrium is established.

(i) State **two** features of a reaction that is in dynamic equilibrium.

1

2

[2]

(ii) The equilibrium constant, K_p , is given by the following expression.

$$K_p = \frac{p_{\text{O}_2}}{p_{\text{Cl}_2}^2}$$

At $1.00 \times 10^5 \text{ Pa}$ and 500 K , 70% of the initial amount of $\text{Cl}_2\text{(g)}$ has reacted.

Calculate K_p and state its units.

$$K_p =$$

units =

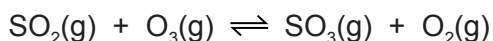
[3]

39. O/N 19/P22/Q1/d

1 In the Periodic Table, the p block contains elements whose outer electrons are found in the p subshell.

(d) SO_2 can react with ozone, O_3 , to form SO_3 in two different reactions.

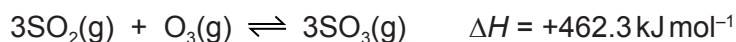
(i) In one reaction, SO_2 reacts with O_3 until a dynamic equilibrium is established.



State and explain the effect of an increase in pressure on the composition of the equilibrium mixture.

[2]

(ii) In the other reaction, a different equilibrium is established at 300 K as shown.



Suggest a temperature needed to increase the yield of SO_3 at equilibrium.

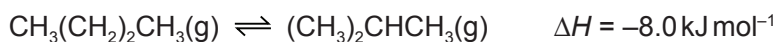
Explain your answer.

[2]

40. M/J 19/P22/Q1/b,c

1 Methylpropane, $(\text{CH}_3)_2\text{CHCH}_3$, is an isomer of butane, $\text{CH}_3(\text{CH}_2)_2\text{CH}_3$.

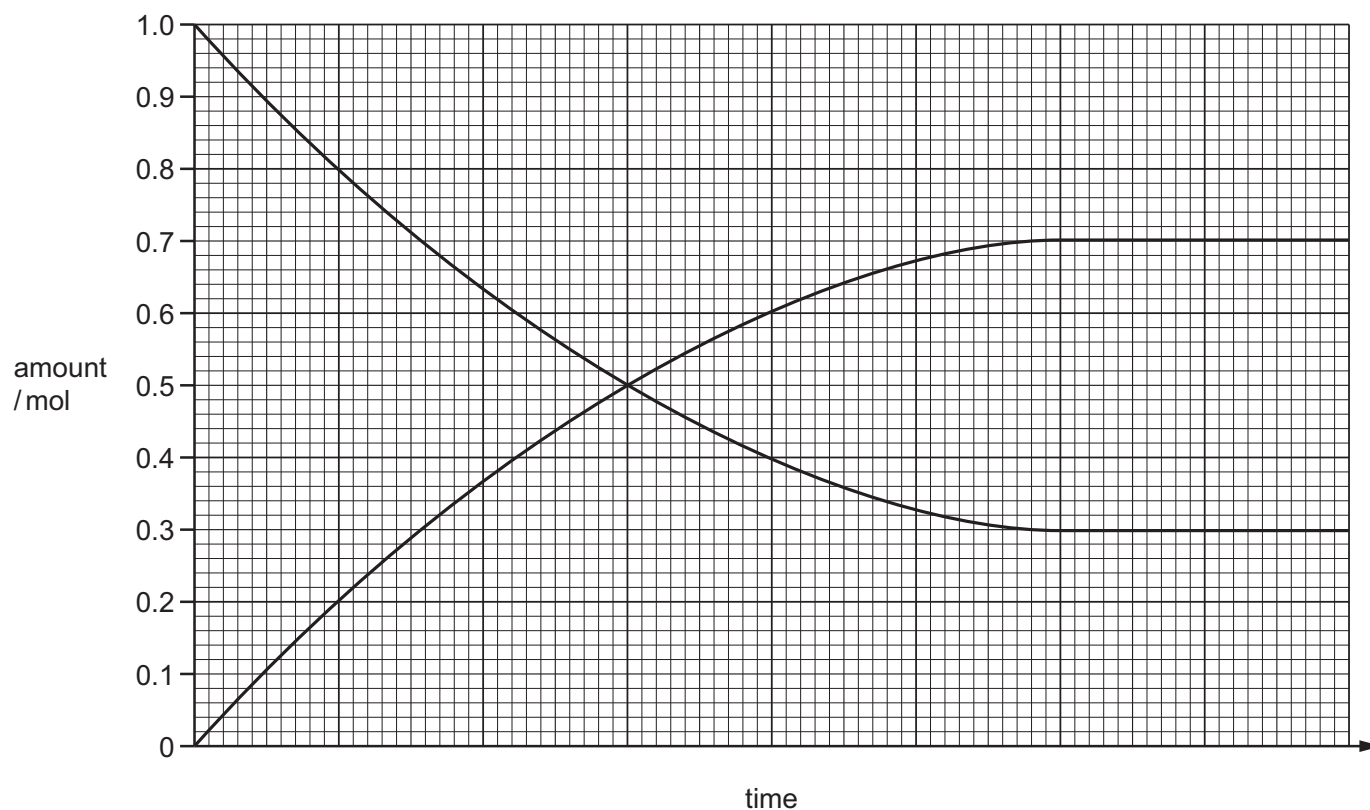
(b) When a sample of butane is heated to 373 K, in the presence of a catalyst, and allowed to reach equilibrium the following reaction occurs.



State and explain the effect on the composition of this equilibrium mixture when the temperature is increased to 473 K.

[2]

(c) 1 mole of butane gas was added to a 1 dm^3 closed system, at a constant temperature and pressure. The amount of butane and methylpropane was measured at regular time intervals.



(i) Label the graph with a t to show the time taken to reach dynamic equilibrium. [1]

(ii) Use the graph to find the concentration of butane and methylpropane in the mixture at equilibrium.

concentration of butane = mol dm^{-3}

concentration of methylpropane = mol dm^{-3}

[1]

(iii) Write an expression for K_c for this reaction.

[1]

(iv) Calculate a value for K_c and state its units.

$K_c =$

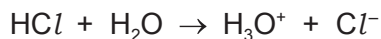
units =

[2]

41. M/J 19/P22/Q2/c(i,ii)

2 Group 17 elements are commonly referred to as the halogens.

(c) Hydrogen chloride reacts with water.



(i) In this reaction, one of the reactants behaves as a Brønsted-Lowry acid.

What is meant by the term *Brønsted-Lowry acid*?

[1]

(ii) Identify the reactant behaving as an acid and its conjugate base.

acid

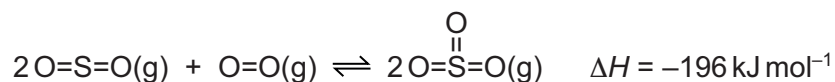
conjugate base

[1]

42. M/J 18/P21/Q1/d(ii),f(ii,iii)

1 Sulfuric acid is manufactured by the Contact process.

One stage in this process is the conversion of sulfur dioxide into sulfur trioxide in the presence of a heterogeneous catalyst of vanadium(V) oxide, V_2O_5 .



(ii) State and explain the effect of increasing temperature on the yield of SO_3 .

[3]

(f) SO_2 reacts with water to form sulfurous acid.

Sulfurous acid is a weak Brønsted-Lowry acid, while sulfuric acid is a strong Brønsted-Lowry acid.

(ii) State the meaning of the term *strong Brønsted-Lowry acid*.

[2]

(iii) Write an equation to show the acid-base behaviour of sulfuric acid with water. Include state symbols.

[2]

43. M/J 18/P22/Q2/c

2 Ammonium iron(II) sulfate, $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2$, has a relative formula mass, M_r , of 284.

(c) In aqueous solution the ammonium ion acts as a weak Brønsted-Lowry acid.

(i) Explain the meaning of the term *weak Brønsted-Lowry acid*.

[2]

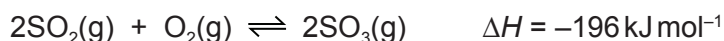
(ii) Write an equation to show this behaviour of the ammonium ion in water. Include state symbols.

[2]

44. M/J 18/P23/Q1/c(i,iii),d

1 The elements sodium to chlorine, in the third period, all form oxides.

(c) SO_3 is produced by the reaction between SO_2 and O_2 in the Contact process. A dynamic equilibrium is established.



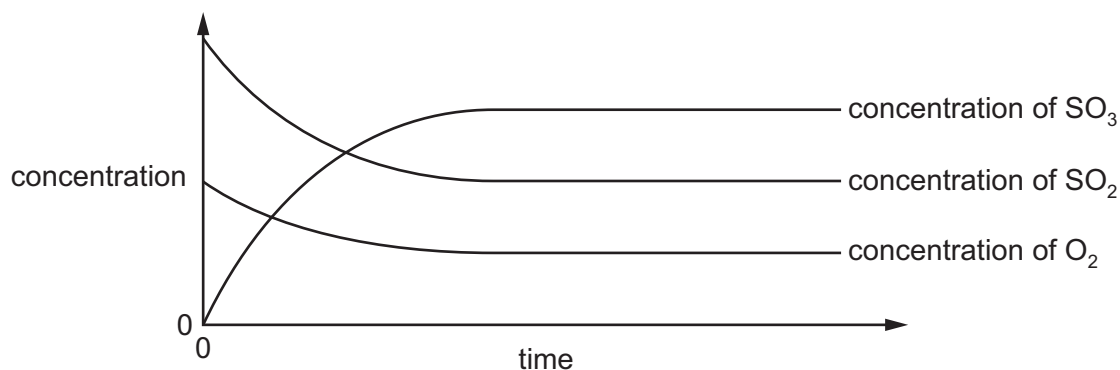
(i) Explain why increasing the total pressure, at constant temperature, increases the rate of production of SO_3 and increases the yield of SO_3 .

rate

yield

[4]

The graph shows how the concentrations of all three species in the system change with time for a typical reaction mixture. The gradients of all three lines decrease with time and then level off in this dynamic equilibrium.

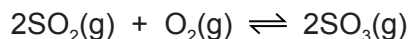


(iii) Explain why all three lines become horizontal.

[1]

- (d) 2.00 moles of $\text{SO}_2(\text{g})$ and 2.00 moles of $\text{O}_2(\text{g})$ are sealed in a container with a suitable catalyst, at constant temperature and pressure. The resulting equilibrium mixture contains 1.98 moles of $\text{SO}_3(\text{g})$.

The total volume of the equilibrium mixture is 40.0 dm^3 .



- (i) Write the expression for the equilibrium constant, K_c , for the reaction between $\text{SO}_2(\text{g})$ and $\text{O}_2(\text{g})$ to produce $\text{SO}_3(\text{g})$.

$K_c =$

[1]

- (ii) Calculate the amount, in moles, of $\text{SO}_2(\text{g})$ and $\text{O}_2(\text{g})$ in the equilibrium mixture.

$\text{SO}_2(\text{g}) =$ mol

$\text{O}_2(\text{g}) =$ mol
[2]

- (iii) Use your answers to (d)(i) and (d)(ii) to calculate the value of K_c for this equilibrium mixture. Give the units of K_c .

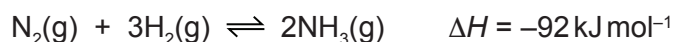
$K_c =$

units =

[3]

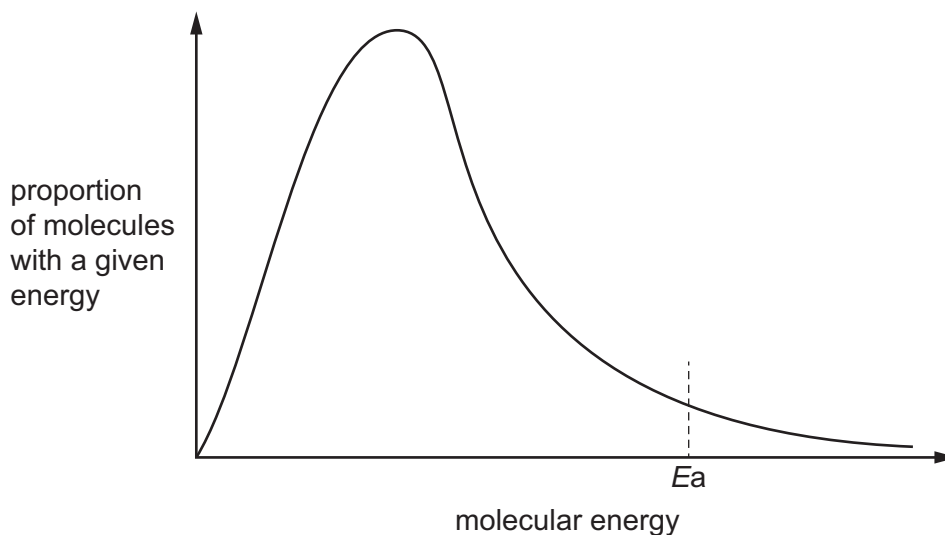
45. O/N 17/P21/Q1/b(iii),c,d

- 1 Ammonia, NH_3 , is manufactured from nitrogen and hydrogen by the Haber process.



- (b) The Haber process is usually carried out at a temperature of approximately 400°C in the presence of a catalyst. Changing the temperature affects both the rate of production of ammonia and the yield of ammonia.

The Boltzmann distribution for a mixture of nitrogen and hydrogen at 400°C is shown. E_a represents the activation energy for the reaction.



- (iii) State and explain the effect of increasing temperature on the yield of ammonia.
Use Le Chatelier's principle to explain your answer.

[3]

- (c) At a pressure of 2.00×10^7 Pa, 1.00 mol of nitrogen, $\text{N}_2(\text{g})$, was mixed with 3.00 mol of hydrogen, $\text{H}_2(\text{g})$. The final equilibrium mixture formed contained 0.300 mol of ammonia, $\text{NH}_3(\text{g})$.

- (i) Calculate the amounts, in mol, of $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$ in the equilibrium mixture.

$\text{N}_2(\text{g}) =$ mol

$\text{H}_2(\text{g}) =$ mol
[2]

- (ii) Calculate the partial pressure of ammonia, p_{NH_3} , in the equilibrium mixture.

Give your answer to **three** significant figures.

$p_{\text{NH}_3} =$ Pa [3]

- (d) In another equilibrium mixture the partial pressures are as shown.

substance	partial pressure / Pa
$\text{N}_2(\text{g})$	2.20×10^6
$\text{H}_2(\text{g})$	9.62×10^5
$\text{NH}_3(\text{g})$	1.40×10^4

- (i) Write the expression for the equilibrium constant, K_p , for the production of ammonia from nitrogen and hydrogen.

$K_p =$

[1]

- (ii) Calculate the value of K_p for this reaction.

State the units.

$K_p =$

units =

[2]

- (iii) This reaction is repeated with the same starting amounts of nitrogen and hydrogen. The same temperature is used but the container has a smaller volume.

State the effects, if any, of this change on the yield of ammonia and on the value of K_p .

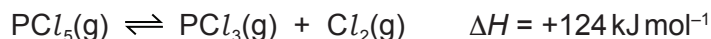
effect on yield of ammonia

effect on value of K_p

[2]

46. O/N 17/P22/Q2/b(ii),c,d

- 2 At 450 K phosphorus(V) chloride, $\text{PCl}_5(\text{g})$, decomposes to form phosphorus(III) chloride, $\text{PCl}_3(\text{g})$, and chlorine, $\text{Cl}_2(\text{g})$. A dynamic equilibrium is established as shown.



- (ii) State and explain the effect of increasing temperature on the percentage of $\text{PCl}_5(\text{g})$ that decomposes.

[2]

- (c) Explain the meaning of the term *dynamic equilibrium* and the conditions necessary for it to become established.

[2]

- (d) When 2.00 mol of $\text{PCl}_5(\text{g})$ are decomposed at 450 K and $1.00 \times 10^5 \text{ Pa}$ the resulting equilibrium mixture contains 0.800 mol of $\text{Cl}_2(\text{g})$.

- (i) Calculate the partial pressure of phosphorus(V) chloride, $p\text{PCl}_5$, in this equilibrium mixture.

$$p\text{PCl}_5 = \quad \text{Pa} \quad [2]$$

- (ii) Write the expression for the equilibrium constant, K_p , for the decomposition of $\text{PCl}_5(\text{g})$.

$$K_p =$$

[1]

- (iii) The partial pressures of $\text{PCl}_3(\text{g})$ and of $\text{Cl}_2(\text{g})$ in this equilibrium mixture are both $2.86 \times 10^4 \text{ Pa}$.

Calculate the value of K_p and state its units.

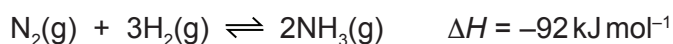
$$K_p =$$

$$\text{units} =$$

[2]

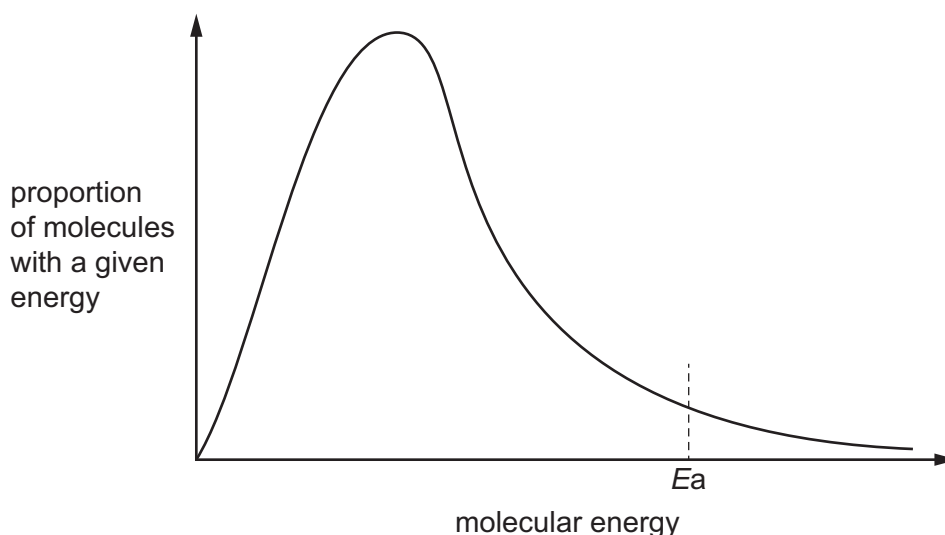
47. O/N 17/P23/Q1/b(iii),c,d

- 1 Ammonia, NH_3 , is manufactured from nitrogen and hydrogen by the Haber process.



- (b) The Haber process is usually carried out at a temperature of approximately 400°C in the presence of a catalyst. Changing the temperature affects both the rate of production of ammonia and the yield of ammonia.

The Boltzmann distribution for a mixture of nitrogen and hydrogen at 400°C is shown.
 E_a represents the activation energy for the reaction.



- (iii) State and explain the effect of increasing temperature on the yield of ammonia. Use Le Chatelier's principle to explain your answer.

[3]

- (c) At a pressure of 2.00×10^7 Pa, 1.00 mol of nitrogen, $\text{N}_2(\text{g})$, was mixed with 3.00 mol of hydrogen, $\text{H}_2(\text{g})$. The final equilibrium mixture formed contained 0.300 mol of ammonia, $\text{NH}_3(\text{g})$.

- (i) Calculate the amounts, in mol, of $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$ in the equilibrium mixture.

$\text{N}_2(\text{g}) =$ mol

$\text{H}_2(\text{g}) =$ mol

[2]

- (ii) Calculate the partial pressure of ammonia, $p\text{NH}_3$, in the equilibrium mixture.

Give your answer to **three** significant figures.

$p\text{NH}_3 =$ Pa [3]

- (d) In another equilibrium mixture the partial pressures are as shown.

substance	partial pressure / Pa
$\text{N}_2(\text{g})$	2.20×10^6
$\text{H}_2(\text{g})$	9.62×10^5
$\text{NH}_3(\text{g})$	1.40×10^4

- (i) Write the expression for the equilibrium constant, K_p , for the production of ammonia from nitrogen and hydrogen.

$K_p =$

[1]

- (ii) Calculate the value of K_p for this reaction.

State the units.

$K_p =$

units =

[2]

- (iii) This reaction is repeated with the same starting amounts of nitrogen and hydrogen. The same temperature is used but the container has a smaller volume.

State the effects, if any, of this change on the yield of ammonia and on the value of K_p .

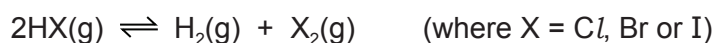
effect on yield of ammonia

effect on value of K_p

[2]

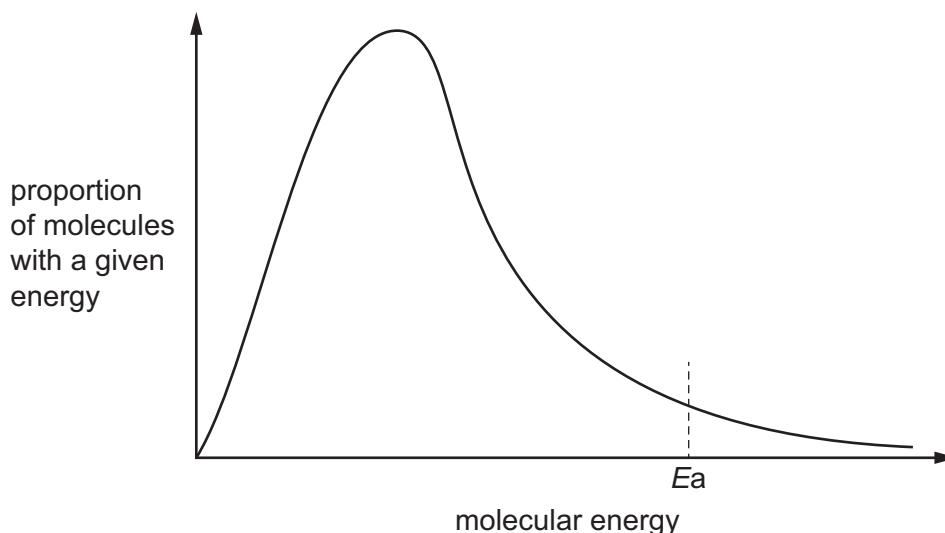
48. M/J 17/P22/Q3/b(iii),c,d

- 3 The hydrogen halides, HCl, HBr and HI, can undergo thermal decomposition. In a sealed container an equilibrium is established according to the equation shown.



- (b) At a temperature of 700 K a sample of HBr is approximately 10% decomposed. Changing the temperature affects both the rate of decomposition of HBr and the percentage that decomposes.

The Boltzmann distribution for a sample of HBr at 700 K is shown. E_a represents the activation energy for the reaction.



- (iii) The decomposition of HBr is endothermic.

State the effect of increasing temperature on the **percentage** of HBr that decomposes. Use Le Chatelier's principle to explain your answer.

[3]

- (c) At temperatures above 1500 K, HCl will decompose.

A sample of 0.300 mol of HCl decomposed in a sealed container.

The resulting equilibrium mixture was found to contain 1.50×10^{-2} mol of Cl_2 .

- (i) Calculate the amounts, in mol, of H_2 and HCl present in the equilibrium mixture.

$\text{H}_2 =$

mol

HCl = mol
[2]

(ii) Calculate the mole fraction of each gas in the equilibrium mixture.

mole fraction of HCl =

mole fraction of H₂ =

mole fraction of Cl₂ =

[1]

(d) In another experiment under different conditions, an equilibrium mixture was produced with mole fractions for each species as shown.

species	mole fraction
HCl	0.88
H ₂	0.06
Cl ₂	0.06

(i) Write the expression for the equilibrium constant, K_p , for the decomposition of HCl.



$K_p =$

[1]

(ii) Explain why the total pressure of the system does **not** need to be known for K_p to be calculated for this experiment.

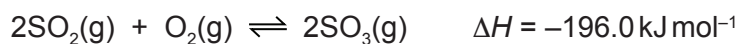
[1]

(iii) Calculate the value of K_p for this experiment.

$K_p =$ [1]

49. M/J 17/P23/Q3/b(iii),c,d

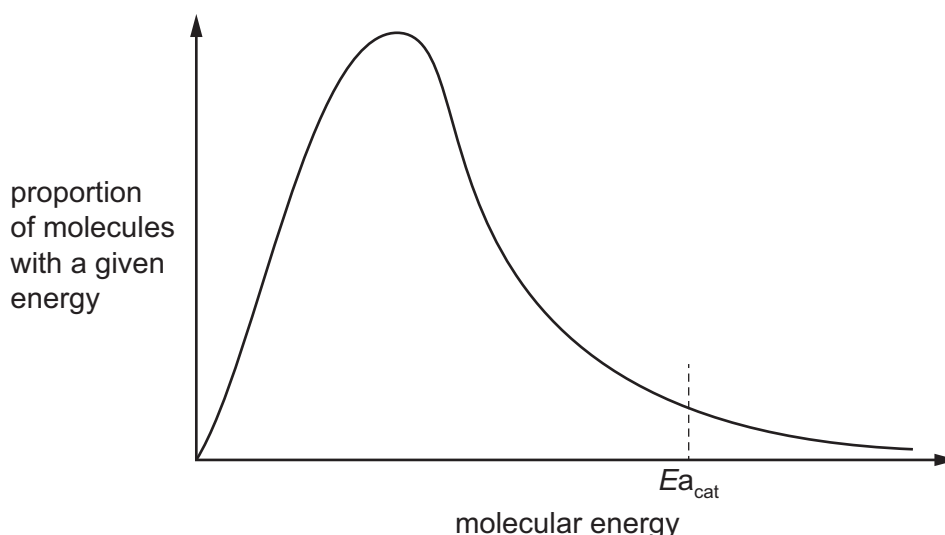
3 Sulfur trioxide, SO₃, is manufactured from sulfur dioxide and oxygen by the Contact process.



(b) The Contact process is usually carried out at a temperature of approximately 700 K, a pressure of approximately 150 kPa and in the presence of a vanadium(V) oxide catalyst, V₂O₅.

The Boltzmann distribution for a mixture of SO₂ and O₂ at 700 K is shown.

$E_{a_{\text{cat}}}$ represents the activation energy for the reaction in the presence of the catalyst.



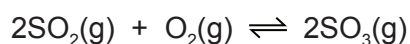
- (iii) State and explain how an increase in pressure would affect both the rate of reaction and the yield of SO_3 in the Contact process.

rate

yield

[4]

- (c) At a pressure of $1.50 \times 10^5 \text{ Pa}$, 1.00 mol of sulfur dioxide gas, SO_2 , was mixed with 1.00 mol of oxygen gas, O_2 . The final equilibrium mixture formed was found to contain 0.505 mol of O_2 .



- (i) Calculate the amount, in mol, of SO_2 and SO_3 in the equilibrium mixture.

$\text{SO}_2 =$ mol

$\text{SO}_3 =$ mol
[1]

- (ii) Calculate the partial pressure of oxygen gas, p_{O_2} , in the equilibrium mixture.

$p_{\text{O}_2} =$ Pa [2]

- (d) In another equilibrium mixture formed from different starting amounts of SO_2 and O_2 , the partial pressures of SO_2 , O_2 and SO_3 were as shown.

$$p_{\text{SO}_2} = 8.42 \times 10^2 \text{ Pa}$$

$$p_{\text{O}_2} = 6.00 \times 10^4 \text{ Pa}$$

$$p_{\text{SO}_3} = 9.10 \times 10^4 \text{ Pa}$$

- (i) Write the expression for the equilibrium constant, K_p , for the production of SO_3 from SO_2 and O_2 .

$K_p =$

[1]

- (ii) Calculate the value of K_p for this reaction and state the units.

$$K_p =$$

units =

[2]

50. MAR 17/P22/Q2/b(i),c(iii,iv,v,vi)

- 2** Hydrogen halides are compounds formed when halogens (Group 17 elements) react with hydrogen. The bond polarity of the hydrogen halides decreases from HF to HI.

Some relevant data are shown in the table.

hydrogen halide	HF	HCl	HBr	HI
boiling point/°C	19	−85	−67	−35
H–X bond energy/kJ mol ^{−1}	562	431	366	299

- (b)** The equation for the preparation of hydrogen chloride using concentrated sulfuric acid is shown.



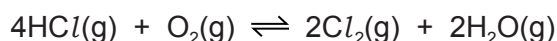
- (i)** Use the Brønsted-Lowry theory of acids and bases to identify the base and its conjugate acid in this reaction. Explain your answer.

Brønsted-Lowry base (base-I) =

conjugate acid (acid-II) =

[2]

- (c)** Hydrogen chloride undergoes a reversible reaction with oxygen.



The reaction is carried out at 400 °C in the presence of a copper(II) chloride catalyst.

- (iii)** The reaction exists in dynamic equilibrium.

The reaction was repeated at 1000 °C and the same pressure.

State and explain the effect on the composition of the equilibrium mixture of the change in temperature.

[2]

- (iv)** When 1.60 mol of HCl are mixed in a sealed container with 0.500 mol of O₂ at 400 °C, 0.600 mol of Cl₂ and 0.600 mol of H₂O are formed.

The total pressure inside the container is 1.50 × 10⁵ Pa.

- Calculate the amounts, in mol, of HCl and O₂ in the equilibrium mixture.

HCl = _____ mol

O₂ = _____ mol

- Calculate the mole fraction of Cl₂ and hence the partial pressure of Cl₂ in the equilibrium mixture.

mole fraction of $\text{Cl}_2 =$
 $p_{\text{Cl}_2} =$ Pa
[3]

- (v) In a separate experiment, an equilibrium reaction mixture was found to contain the four gases at the partial pressures shown in the table.

gas	HCl	O ₂	Cl ₂	H ₂ O
partial pressure / Pa	4.8×10^4	3.0×10^4	3.6×10^4	3.6×10^4

$$K_p = \frac{(p_{\text{Cl}_2})^2 \times (p_{\text{H}_2\text{O}})^2}{(p_{\text{HCl}})^4 \times p_{\text{O}_2}}$$

Use this information and the expression given for K_p to calculate a value for K_p . State the units of K_p .

 $K_p =$
units =
[2]

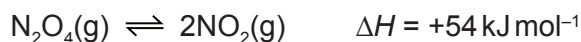
- (vi) The reaction is repeated without a catalyst.

State the effect of this on K_p .

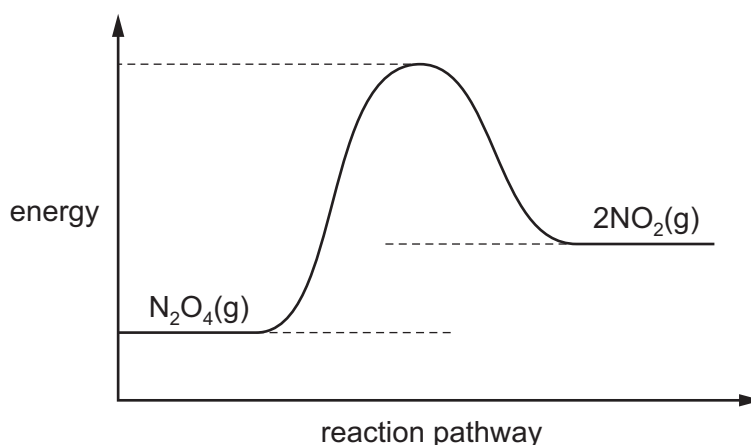
[1]

51. O/N 16/P22/Q2/b(ii,iii,iv,v,vi,vii)

- 2 Dinitrogen tetroxide, N_2O_4 , and nitrogen dioxide, NO_2 , exist in dynamic equilibrium with each other.



The energy profile for this reaction is shown.



- (b) 0.0500 mol of N_2O_4 was placed in a sealed vessel of volume 1.00 dm^3 , at a temperature of 50°C and a pressure of $1.68 \times 10^5 \text{ Pa}$. The mass of the resulting equilibrium mixture was 4.606 g .

- (ii) The number of moles of N_2O_4 that dissociated can be represented by n .

State, in terms of n , the amount, in moles, of NO_2 in the equilibrium mixture.

moles of $\text{NO}_2 =$ [1]

The number of moles of N_2O_4 remaining at equilibrium is $(0.05 - n)$.

(iii) State, in terms of n , the total amount, in moles, of gas in the equilibrium mixture.

[1]

(iv) State, in terms of n , the mole fraction of NO_2 in the equilibrium mixture.

[1]

In this equilibrium mixture, the mole fraction of NO_2 is 0.400.

(v) Use your answers to (ii) and (iv) to calculate the amount in moles of each gas in the equilibrium mixture. Give your answers to **three** significant figures.

amount of N_2O_4 = mol

amount of NO_2 = mol
[2]

(vi) Write the expression for the equilibrium constant, K_p , for this equilibrium.

K_p =

[1]

(vii) Use the total pressure of the mixture, $1.68 \times 10^5 \text{ Pa}$, to calculate the value of the equilibrium constant, K_p , and give its units.

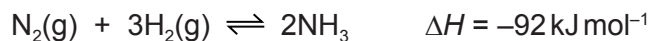
K_p =

units =

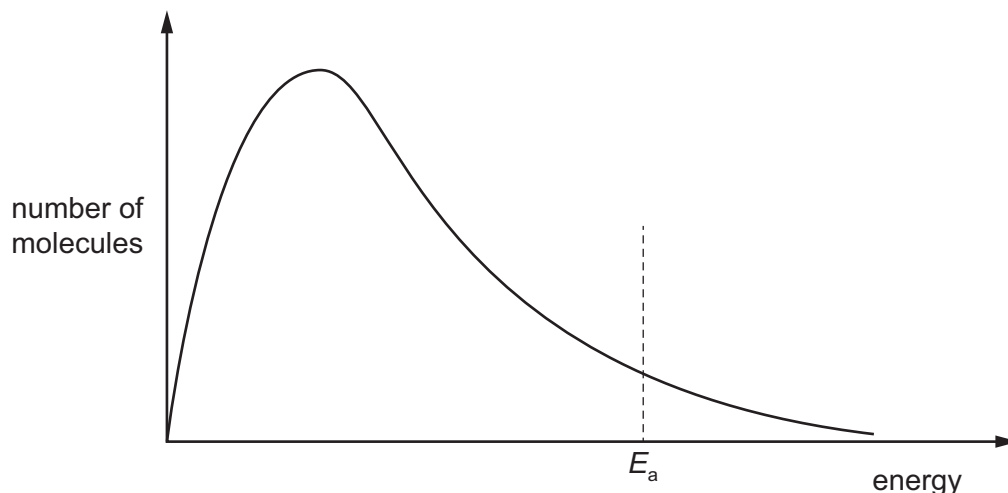
[3]

52. M/J 16/P23/Q2/b(iv)

- 2 Ammonium nitrate is an important fertiliser made by the acid-base reaction between ammonia and nitric acid.
- (b) The Haber process involves a reaction between nitrogen and hydrogen at a temperature of 450 °C and a pressure of 20 000 kPa. At a higher reaction temperature, the rate of production of ammonia would be greater.



The Boltzmann distribution curve shows the distribution of energies in a mixture of nitrogen and hydrogen at 450 °C.



- (iv) Explain why a higher temperature is **not** used despite the fact that it would increase the rate of production of ammonia.

[2]

53. MAR 16/P22/Q3/b

- 3 Over one million tonnes of hydrogen cyanide, HCN, are produced each year using the Andrussow process. The overall equation for the reaction is shown.



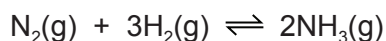
- (b) The reaction exists as a dynamic equilibrium.
- (i) Explain what is meant by the term *dynamic equilibrium*.
- (ii) State and explain how the amounts of the chemicals present in the equilibrium mixture will change when the pressure is increased.

[1]

[2]

54. O/N 15/P21/Q2/b(ii,iii)

(b) Ammonia is manufactured by the Haber process.



(ii) State the essential operating conditions for the Haber process.

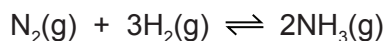
[3]

(iii) Explain the choices of temperature and pressure for the Haber process.

[4]

55. O/N 15/P23/Q2/b(ii,iii)

(b) Ammonia is manufactured by the Haber process.



(ii) State the essential operating conditions for the Haber process.

[3]

(iii) Explain the choices of temperature and pressure for the Haber process.

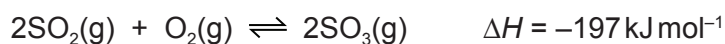
[4]

56. M/J 15/P22/Q2/b

2 Sulfuric acid is an important chemical with a variety of uses.

It is manufactured by the Contact process, the first stage of which involves the conversion of sulfur or a sulfide ore, such as galena, PbS , into sulfur dioxide, SO_2 .

(b) The second stage of the Contact process involves the production of sulfur trioxide, SO_3 , from sulfur dioxide.



(i) State the temperature usually chosen for this conversion and explain this in terms of reaction rates and Le Chatelier's principle.

temperature

explanation

[3]

(ii) State and explain the pressure conditions that would give the best rate and best yield of sulfur trioxide. Explain why these conditions are **not** actually used.

[3]

Answers Section

1. M/J 26/P21/Q2/b(i)

2(b)(i)	increases yield AND <u>equilibrium</u> shifts to right (hand side)	1 A alternatives to increased yield = increased amount / more product / SO_3 A <u>equilibrium</u> shifts to the side with fewer gaseous molecules / to side with fewer gaseous moles A <u>equilibrium</u> shifts to the right to reduce the concentration of SO_2 BOD <u>equilibrium</u> shifts to (favour) forwards reaction / direction I equilibrium shifts to side with lower volume I limiting reagent I any comments on rate R equilibrium shifts to side with more gaseous moles
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2. M/J 26/P22/Q1/e(i)

1(e)(i)	M1 high pressure OR ≥ 200 atm M2 iron catalyst / Fe catalyst	2 A answers on either line M1 BOD equivalent pressure of ≥ 150 atm or equivalent in alternative units with appropriate conversion M1 A $\geq 15\,000$ kPa / ≥ 150 bar M1 I ATP M1 I numbers with no units M2 A alternative correct named catalysts e.g. Co (consult TL if necessary) M2 BOD Fe(II) catalyst / Fe^{2+} catalyst / Fe^{3+} catalyst / iron oxide catalyst M2 BOD magnetite catalyst M2 I Ni M2 I promoters M2 R Pt / platinum / vanadium oxide
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3. M/J 26/P22/Q3/a

3(a)(i)	$\text{H}_2\text{SO}_4 + \text{NaCl} \rightarrow \text{NaHSO}_4 + \text{HCl}$ OR $\text{H}_2\text{SO}_4 + 2\text{NaCl} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{HCl}$	1 A $\text{Cl}^- + \text{H}_2\text{SO}_4 \rightarrow \text{HCl} + \text{HSO}_4^-$ A $2\text{Cl}^- + \text{H}_2\text{SO}_4 \rightarrow 2\text{HCl} + \text{SO}_4^{2-}$
3(a)(ii)	M1 H_2SO_4 acid AND Cl^- base M2 H_2SO_4 donates proton(s) AND Cl^- accepts protons OR H_2SO_4 gives / transfers protons <u>to</u> Cl^-	2 proton = hydrogen ion I general definition of BL acids and bases. $\text{Cl}^- = \text{Cl}$ in NaCl I HCl is base/protons given to HCl Alternative if neither M1 nor M2 can be awarded: H_2SO_4 acid Cl^- base H_2SO_4 donates proton(s) Cl^- accepts proton(s) •✓•✓

4. M/J 26/P23/Q3/b,c,d

3(b)	M1 (if a change in conditions occurs) the equilibrium shifts M2 to minimise the change (in conditions)	2 M1 A equilibrium moves / equilibrium changes M1 I there will be a 'change in the system' M2 minimise = oppose = counteract = resists = reverse
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5. M/J 26/P24/Q2/b(ii),c

2(b)(ii)	M1 (temperature:) 400–450 °C M2 (pressure:) 150–250 atm	2 Units required M1 A alternative temperatures with appropriate units A 400–500 °C A 673–773 K M2 A 15 200–25 300 kPa A 15 200 000–25 300 000 Pa A 150–250 bar
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2(c)(i)	<table><tr><td></td><td>N₂(g)</td><td>H₂(g)</td><td>NH₃(g)</td></tr><tr><td>initial amount / mol</td><td>3.55</td><td>8.60</td><td>0</td></tr><tr><td>change in amount / mol</td><td>(-) 1.62</td><td>(-) 4.86</td><td>3.24</td></tr><tr><td>amount at equilibrium / mol</td><td>(+) 1.93</td><td>(+) 3.74</td><td>3.24</td></tr></table>		N ₂ (g)	H ₂ (g)	NH ₃ (g)	initial amount / mol	3.55	8.60	0	change in amount / mol	(-) 1.62	(-) 4.86	3.24	amount at equilibrium / mol	(+) 1.93	(+) 3.74	3.24	3 M1 (-) 1.62 M2 (-) 4.86 M3 N ₂ and H ₂ amount / moles at equilibrium (initial amount / moles minus change in amount / moles) M3 A ecf from incorrect changes in amount / moles in M1 / M2 M3 R negative amounts / moles at equilibrium Max 2: <table><tr><td></td><td>N₂(g)</td><td>H₂(g)</td><td>NH₃(g)</td></tr><tr><td>initial amount /mol</td><td>3.55</td><td>8.60</td><td>0</td></tr><tr><td>change in amount /mol</td><td>(-)1.93</td><td>(-)3.74</td><td>3.24</td></tr><tr><td>amount at equilibrium /mol</td><td>(+)1.62</td><td>(+)4.86</td><td>3.24</td></tr></table>		N ₂ (g)	H ₂ (g)	NH ₃ (g)	initial amount /mol	3.55	8.60	0	change in amount /mol	(-)1.93	(-)3.74	3.24	amount at equilibrium /mol	(+)1.62	(+)4.86	3.24
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amount at equilibrium /mol	(+)1.62	(+)4.86	3.24																															
2(c)(ii)	M1 Total amount / mol = 2.561 OR 0.874 + 1.340 + 0.347 M2 Mole fraction of NH ₃ = 0.347 ÷ 2.561 = 0.135	2 R M1 and M2 if 0.874 + 1.340 + 0.347 is not used to calculate total amount / moles e.g. total moles = 0.874 + 1.340 M2 A ecf if 0.874 + 1.340 + 0.347 are used to calculate total amount / moles, but incorrectly e.g. 0.874 + 1.340 + 0.347 ² A correctly rounded to minimum of 2 significant figures A as integer fraction e.g. 347 / 2561 I 13.5% A calculator answer 0.135494																																
2(c)(iii)	(partial pressure) = mole fraction (of a gas) × total pressure OR the pressure that would be exerted by (only) one of the gases in a mixture (of gases) (if it occupied the same volume)	1 A product of mole fraction and total pressure A the pressure that is exerted by one of the gases to the total pressure A The pressure that a gas in a mixture of gases exerts on the surroundings A ratio / proportion / fraction of the pressure of a gas to the total pressure (in a closed system) A gas = molecules = particles R compound / element I amount of pressure that a particular gas occupies compared to total pressure																																
2(c)(iv)	M1 $p(\text{NH}_3)^2 = K_p \times p(\text{N}_2) \times p(\text{H}_2)^3$ OR $p(\text{NH}_3)^2 = 8.49 \times 10^{-7} \times 1590 \times (1750)^3$ OR $p(\text{NH}_3) = \sqrt{8.49 \times 10^{-7} \times 1590 \times (1750)^3}$ M2 $p(\text{NH}_3)^2 = 7.23(4673906) \times 10^6$ M3 $\sqrt{\text{M2}} = 2690 / 2689.73$	3 M2 ecf from.... M1 transcription error or M1 incorrect rearrangement for $p(\text{NH}_3)^2$ using all the terms K_p AND $p(\text{N}_2)$ AND $p(\text{H}_2)^3$ M2 A calculator answer 723 4673.906 M3 ecf $\sqrt{\text{M2}}$ correctly calculated to minimum 3 significant figures M3 A calculator answer 2689.73																																

6. MAR 26/P22/Q4/a(i,ii,iii)

4(a)(i)	(proton donor that) partially dissociates (in aqueous solution)	1
4(a)(ii)	M1 weak acid curve starts pH 1.7 (calculation not necessary) M2 vertical section at 25 cm ³ M3 strong alkali curve ends at pH 13	3
4(a)(iii)	$\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$	1

7. O/N 25/P22/Q2/e

2(e)(i)	rate of forward and backward (reaction) are the same / equal OR (ratio of) concentrations of reactants and products are constant	1
2(e)(ii)	closed system	1
2(e)(iii)	M1 (mixture / solution becomes more) yellow M2 (forward) reaction is exothermic (equilibrium shifts left) OR equilibrium / reaction shifts (left) to absorb (additional) heat / energy M3 (mixture / solution becomes more) red M4 fewer moles of gas on right-hand side OR (gas on left and) no gas on the right	4

8. O/N 25/P24/Q2/b

2(b)(i)	mol ⁻² dm ⁶	1
2(b)(ii)	M1 at equilibrium: moles of CO = 0.120 AND moles of H ₂ = 0.240 M2 [CH ₃ OH] = 0.056 AND [CO] = 0.024 AND [H ₂] = 0.048 M3 use of K_c with correct evaluation = 1.01×10^3	3
2(b)(iii)	no effect AND K_c is not affected by pressure	1

9. M/J 25/P21/Q3/b,c

3(b)	M1 (if a change is made at a system is dynamic equilibrium) the equilibrium moves M2 to minimise the change in conditions	2
3(c)	(forward) reaction is exothermic (a + b) > c	2

10. M/J 25/P23/Q2/d

2(d)	it / ZnO accepts protons	1
------	--------------------------	----------

11. M/J 25/P24/Q3/b,c

3(b)	M1 $0.2 - 0.154 = 0.046$ (mol H ₂ and / or I ₂ reacts) M2 mol H ₂ at equilibrium $0.1 - 0.046 = 0.054$ M3 mol HI at equilibrium $0.3 + (0.046 \times 2) = 0.392$	3
3(c)	2.46	1

12. O/N 24/P21/Q3/b(iii)

3(b)(iii)	M1 NH ₄ ⁺ / ammonium ion / ammonium cation is a proton / H ⁺ donor M2 OH ⁻ is a proton / H ⁺ acceptor OR the OH ⁻ / NaOH is a stronger base than NH ₃	2
-----------	--	----------

13. O/N 24/P23/Q3/b(iii)

3(b)(iii)	M1 NH_4^+ / ammonium ion / ammonium cation is a proton / H^+ donor M2 OH^- is a proton / H^+ acceptor OR the OH^- / NaOH is a stronger base than NH_3	2
-----------	---	----------

14. M/J 24/P21/Q2/c

2(c)(i)			oxidation number in Mg ₃ N ₂	Both correct for one mark	1
		magnesium	(+)II / 2(+)		
		nitrogen	–III/ 3 –		
2(c)(ii)	Mg ₃ N ₂ + 6H ₂ O → 3Mg(OH) ₂ + 2NH ₃				1
2(c)(iii)	M1 due to OH [–] (aq) M2 EITHER acid-base reaction of water by ammonia OR dissociation of (some) Mg(OH) ₂				2

15. M/J 24/P21/Q3/a,b,c

3(a)	M1 (if a change in conditions occurs) the equilibrium shifts M2 to minimise the change in conditions	2
3(b)	M1 change in appearance paler red / more yellow / less orange M2 change in relative concentration of $\text{FeSCN}^{2+}(\text{aq})$ lower M3 change in value of the equilibrium constant, K_c lower	3
3(c)(i)	M1 deduce the concentration of SCN^- present in initial mixture $[\text{SCN}^- \text{ eq'm}] = [\text{Fe}^{3+} \text{ eq'm}] = [\text{original } \text{SCN}^-] - [\text{FeSCN}^{2+} \text{ produced}]$ $1.30 \times 10^{-3} + 0.300 \times 10^{-3} = 1.60 \times 10^{-3} \text{ mol dm}^{-3}$ M2 find initial amount of Fe^{3+} initially $\text{M1} \times 25 / 1000 = \text{amount of } \text{Fe}^{3+} \text{ added initially} = 4.00 \times 10^{-5} \text{ (mol)}$	2
3(c)(ii)	M1 value for $K_c = 178$ M2 units = $\text{mol}^{-1} \text{ dm}^3$	2

16. M/J 24/P22/Q3/a,b

3(a)	M1 rates of forward and reverse / backward reactions are equal M2 no change in measurable properties OR concentration of reactants AND products remain constant	2
3(b)(i)	M1 change to appearance of the mixture: (goes) darker red / more red M2 change to relative concentration of $\text{Fe}^{3+}(\text{aq})$: decreases M3 change to the value of equilibrium constant, K_c : constant / none	3
3(b)(ii)	Method 1 M1 conc of initial conc of Fe^{3+} and $\text{SCN}^- = 0.0020 \text{ (mol dm}^{-3}\text{)}$ M2 at equilibrium the conc of Fe^{3+} and SCN^- are ($\text{M1} - 0.000423$) M3 use $0.000423 \div (\text{M1} - 0.000423)^2$ to calculate $K_c = 170(.089)$ M4 units = $\text{mol}^{-1} \text{ dm}^3$ OR $\text{dm}^3 \text{ mol}^{-1}$ Method 2 M1 no mol $\text{FeSCN}^{2+} = 4.23 \times 10^{-4} \times 25 / 1000 = 1.0575 \times 10^{-5}$ M2 no mol Fe^{3+} and SCN^- are ($5 \times 10^{-5} - \text{M1}$) = 3.9425×10^{-5} M3 $0.000423 \div (\text{M2} \times 1000 / 25)^2 = 170(.089)$ M4 units = $\text{mol}^{-1} \text{ dm}^3$ OR $\text{dm}^3 \text{ mol}^{-1}$	4

17. M/J 24/P23/Q2/a,b(i),c

2(a)	58% / no change / stays the same AND catalysts don't affect yield / iron is a catalyst	1
2(b)(i)	less yield (in C than A) AND (as) exothermic (forward reaction)	1

2(c)(i)	M1 $K_p = (p\text{NH}_3)^2 / (p\text{N}_2)(p\text{H}_2)^3$ M2 atm^{-2}	2
2(c)(ii)	M1 correct amount N_2 at equilibrium = $1 - 1.16 / 2 = 0.42$ AND correct amount H_2 at equilibrium = $3.0 - (1.16 \times 3 / 2) = 1.26$ M2 express amount of ammonia as a mol fraction in terms of total amount of $\text{N}_2 + \text{H}_2 + \text{NH}_3$ at equilibrium = $1.16 / (\text{N}_2 + \text{H}_2 + 1.16) = 0.408$ OR 0.41	2
2(c)(iii)	$0.625 \times 1000 = 625$ (atm)	1

18. M/J 24/P23/Q4/b(i)

4(b)(i)	acid–base reaction	1
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19. O/N 23/P22/Q2/a,b,c

2(a)	curved line from (0,0) to reach same horizontal line	1
2(b)	equilibrium position would / it moves to left / reactants	1
	forward reaction is exothermic (so disfavoured by increase in T)	1
2(c)	$\frac{0.605}{1.60}$ NO $(= 0.378125)$ $\frac{0.605}{1.60}$ NO_2 $(= 0.378125)$ $\frac{0.390}{1.60}$ N_2O_3 $(= 0.24375)$	1
	(partial pressure / kPa 38.19 38.19 24.62) $K_p = 24.62 \div 38.19^2 = 0.0169$	1
	kPa^{-1}	1

20. O/N 23/P22/Q3/a(ii)

3(a)(ii)	proton / H^+ donor	1
	partially dissociates (in solution / in water / when aqueous)	1

21. M/J 23/P22/Q4/b,c(i)

4(b)	M1 amount ethanol increases M2 equilibrium moves to right AND to produce fewer moles gas OR smaller amount of gas (to reduce the pressure)	2
4(c)(i)	acid AND it donates a proton / H^+	1

22. M/J 23/P23/Q4/b,c

4(b)(i)	(from colourless) to purple	1
4(b)(ii)	$K_c = [\text{H}_2][\text{I}_2] / [\text{HI}]^2$ OR $0.0217 = [\text{H}_2][\text{I}_2] / [\text{HI}]^2$	1
4(b)(iii)	M1 $[\text{HI}]$ at equilibrium = $1.7 / 2 = 0.85$ M2 idea that $[\text{H}_2] = [\text{I}_2]$ in equilibrium mixture M3 correct calculation based on correct rearrangement of equation in (b)(ii), 0.0217 and their M1 = 0.125	3
4(c)(i)	new mixture will have a lower / smaller conc / amount of H_2 and I_2	1
4(c)(ii)	endothermic AND decrease in temperature causes position of equilibrium to shift in exothermic direction	1

23. MAR 23/P22/Q3/d(ii)

3(d)(ii)	equilibrium moves to right/products where fewer moles / smaller amount of gas	1
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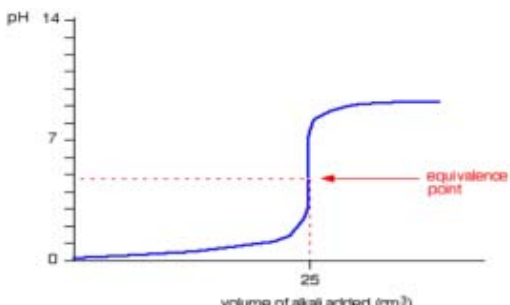
24. O/N 22/P22/Q1/c(i)

1(c)(i)	proton / H ⁺ donor	1
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25. O/N 22/P22/Q2/c(ii)

2(c)(ii)	reaction is exothermic so equilibrium shifts to the LHS OR equilibrium shifts to LHS because forward reaction is exothermic OR equilibrium shifts to LHS because backward reaction is endothermic OR endothermic reaction is favoured so more reactants / are AlCl ₃ is made	1
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26. M/J 22/P21/Q2/d

2(d)(i)	M1 proton / H ⁺ donor M2 fully dissociates (in aqueous solution / water / solvent)	2
2(d)(ii)	NH ₃ + H ₂ O ⇌ NH ₄ ⁺ + OH ⁻	1
2(d)(iii)	M1 correct basic shape extending to ~50 cm ³ with vertical portion of curve at 25 cm ³ M2 initial pH at 0–2 (based on idea that HCl is a strong acid) AND final pH at between 8–12 (based on idea that NH ₃ is a weak alkali) 	2

27. M/J 22/P22/Q3/a,c

3(a)(i)	M1 rate(s) of forward and reverse / backward reactions are equal / are the same M2 no change in measurable properties OR concentration of reactants and products remain constant	2
3(a)(ii)	(colourless gas) becomes purple (gas)	1
3(a)(iii)	$K_p = \frac{(pH_2)(pI_2)}{(pHI)^2}$	1
3(a)(iv)	M1 use correct expression for finding amount of HI used up in reaction 86.4 = 100 × (0.025 – x) / (0.025 – x + x / 2 + x / 2) where x = HI mol used up OR 86.4 = 100 × (0.025 – 2x) / (0.025 – 2x + x + x) where 2x = HI mol used up M2 correct calculation to find amount of HI(mol) at equilibrium based on HI(mol) at eq'm = 0.025 – HI(mol) used = 0.025 – 0.0034 = 0.0216 OR 0.022 (mol) alternative method M1 use the pp H ₂ and pp I ₂ at equilibrium (= 13.6) to calculate no mol H ₂ & I ₂ at equilibrium (either 0.0034 in total OR 0.0017 each) M2 use value for no mol H ₂ + I ₂ in expression to find no mol HI 0.025 – (no mol H ₂ + I ₂) = 0.0216	2
3(c)	none / no change	1

28. M/J 22/P23/Q3/a(iii)

3(a)(iii)	M1 $K_c = \frac{[\text{C}_2\text{H}_6][\text{CO}][\text{CH}_4]}{[\text{C}_4\text{H}_{10}\text{O}]}$ M2 $\text{mol}^2 \text{dm}^{-6}$	2
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29. MAR 22/P22/Q2/d(iii)

2(d)(iii)	position of equilibrium moves / farther to right (at 20 km) [1] (forward) reaction is exothermic AND temperature colder at 20 km (<i>cf.</i> 50 km) [1]	2
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30. MAR 22/P22/Q3/d

3(d)(i)	proton / H^+ donor [1] fully dissociates (in aqueous solution / water / solvent) [1]	2
3(d)(ii)	M1: correct sigmoid shape with vertical section at 25 cm^3 for both M2: both curves show initial $\text{pH} \leq 2$ M3: (with NaOH) heading to $\text{pH} \geq 12$ (with NH_3) heading to $\text{pH} 8-12$ 	3

31. O/N 21/P22/Q1/c,f(iii,iv)

1(c)(i)	$K_p = \frac{p_{\text{HI}}^2}{p_{\text{H}_2} p_{\text{I}_2}}$	1
1(c)(ii)	28.76 OR 28.8 OR 29	1
1(c)(iii)	EITHER option 1 which assumes $\Delta H_{\text{reaction}}$ is (still) endothermic (using the value shown in (b)). M1 (K_p) decreases AND endothermic / ΔH_{f} = + / positive M2 reaction favours formation of reactants / hydrogen and iodine OR (product) yield / partial pressure of HI decreases / equilibrium shifts to the left OR option 2 which realises that $\Delta H_{\text{reaction}}$ is in fact exothermic (using bond energy data in Data Booklet) M1 (K_p) increases AND exothermic / ΔH_{f} = + / negative M2 reaction favours formation of products / hydrogen iodide yield increases / partial pressure of HI increases / equilibrium shifts to the right	2
1(f)(iii)	proton donor / H^+ donor fully dissociates / fully ionises	2
1(f)(iv)	H_2PO_3^-	1

32. M/J 21/P23/Q1/d

1(d)(i)	proton / H^+ acceptor	1
1(d)(ii)	$\text{ClO}^- + \text{H}_2\text{O} \rightarrow \text{HClO} + \text{OH}^-$	1

33. M/J 21/P23/Q2/c

2(c)(i)	M1 rates of forward and reverse / backward reactions are equal	1
	M2 closed / sealed system / container OR no change in measurable properties / no change in macroscopic properties	1

2(c)(ii)	M1 mol fraction = $0.97 / (0.030 + 0.060 + 0.97) = 0.97 / 1.06 = (0.9151)$	1
	M2 $M1 \times 1.0 \times 10^7 = 9.2 \times 10^6 \text{ (Pa)}$ ✓✓	1
2(c)(iii)	<i>expression for K_p</i> M1 $K_p = p(\text{CH}_3\text{OH}) / p(\text{CO}) \times p(\text{H}_2)^2$	1
	M2 units = Pa^{-2}	1

34. MAR 21/P22/Q3/b(i)

3(b)(i)	M1: proton / H^+ donor M2: partially dissociates / does not fully dissociate (in solution)	2
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35. O/N 20/P21/Q2/b(i),d(iii)

2(b)(i)	M1: proton / H^+ donor	2
	M2: partially dissociates (in solution)	
2(d)(iii)	<i>(at 1000 K and 100 kPa)</i> M1: (yield) decreases	4
	M2: reaction is exothermic AND equilibrium moves left	
	<i>(at 500 K and 500 kPa)</i> M3: (yield) increases	
	M4: fewer moles (<u>of gas</u>) on right-hand side AND equilibrium moves right	

36. O/N 20/P22/Q3/d(ii)

3(d)(ii)	M1: proton / H^+ donor	2
	M2: partially dissociates (in solution)	

37. O/N 20/P23/Q2/b(i),d(iii)

2(b)(i)	M1: proton / H^+ donor	2
	M2: partially dissociates (in solution)	
2(d)(iii)	<i>(at 1000 K and 100 kPa)</i> M1: (yield) decreases	4
	M2: reaction is exothermic AND equilibrium moves left	
	<i>(at 500 K and 500 kPa)</i> M3: (yield) increases	
	M4: fewer moles (<u>of gas</u>) on right-hand side AND equilibrium moves right	

38. MAR 20/P22/Q1/a(ii),f

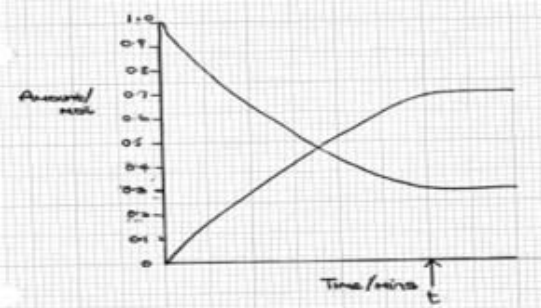
1(a)(ii)	OH^- / hydroxide	1
1(f)(i)	M1 equal rates of forward and backward reactions M2 closed system OR macroscopic properties unchanging	2

1(f)(ii)	M1		$C l_2$	O_2		
		initial	x	0	mol	
		equilibrium	$0.3x$	$0.35x$	mol	
		mol fraction	$\frac{6}{13}$	$\frac{7}{13}$		
	M2	$K_p = \frac{100\,000 \times \frac{7}{13}}{(100\,000 \times \frac{6}{13})^2} = 2.53 \times 10^{-5}$				
		M3 Pa ⁻¹				

39. O/N 19/P22/Q1/d

1(d)(i)	M1: no effect / none M2: equal mol(es) (of gas) on both sides (of equilibrium / equation) owtte	2
1(d)(ii)	M1: (forward reaction is) endothermic M2: Any temperature higher than 300 K	2

40. M/J 19/P22/Q1/b,c

1(b)	(forward reaction is) exothermic reaction	1
	the proportion of methylpropane / product decreases OR the proportion of butane / reactant increases	1
1(c)(i)	t shown on graph which corresponds to start of the horizontal part of both curves. 	1
1(c)(ii)	concentration of butane = 0.3 mol dm^{-3} AND concentration of methylpropane = 0.7 mol dm^{-3}	1
1(c)(iii)	[methylpropane] / [butane] OR $[(CH_3)_2CHCH_3] / [CH_3(CH_2)_2CH_3]$	1
1(c)(iv)	M1 value for K_c $K_c = \frac{\text{value of methylpropane in (ii)}}{\text{value of butane in (ii)}} = 0.7 / 0.3 = 2.3 \text{ (3)}$ M2 units consistent with expression used in M1 no units / dimensionless / none	1

41. M/J 19/P22/Q2/c(i,ii)

2(c)(i)	proton / H ⁺ donor	1
2(c)(ii)	acid HCl AND conjugate base Cl ⁻	1

42. M/J 18/P21/Q1/d(ii),f(ii,iii)

1(d)(ii)	(Increasing T) decreases yield (of SO ₃)	1
	(Forward) reaction is exothermic (or reverse argument)	1
	So increasing T shifts (equilibrium) reaction to left / towards reactants / in endothermic direction (to oppose the change in T)	1
1(f)(ii)	fully ionises/dissociates	1
	(Brønsted-Lowry acid is a) proton / H ⁺ donor	1
1(f)(iii)	H ₂ SO ₄ (l)/(aq) + H ₂ O(l) → HSO ₄ ⁻ (aq) + H ₃ O ⁺ (aq)	
	species and balancing	1
	correct state symbols on left hand side; all products aqueous	1

43. M/J 18/P22/Q2/c

2c(i)	<i>in any order explain meaning of:</i> <i>weak</i> partially ionises / incompletely dissociates (into ions) <i>Bronsted-Lowry acid</i> is a proton donor / H ⁺ (ion) donor / hydrogen ion donor	2
2c(ii)	NH ₄ ⁺ (aq) + H ₂ O(l) ⇌ NH ₃ (aq or g) + H ₃ O ⁺ (aq)	
	all correct species and balancing	1
	correct state symbols	1

44. M/J 18/P23/Q1/c(i,iii),d

1(c)(i)	molecules / particles / reacting species are closer together	1
	so frequency of collisions increases	1
	few(er) moles on right	1
	so (equilibrium) reaction shifts right / towards products / (as pressure increases to oppose the change)	1
1(c)(iii)	(line becomes horizontal when) forward and reverse rates equal / equilibrium established	1
1(d)(i)	$K_c = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2 \times [\text{O}_2]}$	1
1(d)(ii)	SO ₂ = 0.02 (mol)	1
	O ₂ = 1.01 (mol)	1
1(d)(iii)	$K_c = \frac{(1.98 / 40)^2}{(0.02 / 40)^2 (1.01 / 40)}$ $= 3.88 / 3.882 \times 10^5$ Units = dm ³ mol ⁻¹ / mol ⁻¹ dm ³	M1 M2 M3 3

45. O/N 17/P21/Q1/b(iii),c,d

1(b)(iii)	reduces yield (of ammonia).	1
	(increasing T) shifts equilibrium (reaction) to the left / in the reverse direction / towards N_2 and H_2 / towards reactants / in endothermic direction	1
	to oppose the change OR oppose the increase in temperature OR to absorb the (additional) heat / energy OR decrease the temperature	1
1(c)(i)	$N_2 = 0.850$ (mol)	1
	$H_2 = 2.55$ (mol)	1
1(c)(ii)	$n_{TOTAL} = 3.7$ mol	1
	mol fraction of $NH_3 = 0.3 / 3.7$	1
	$p_{NH_3} = 2 \times 10^7 \times (0.3 / 3.7) = 1.62 \times 10^6$	1
1(d)(i)	$K_p = \frac{p_{NH_3}^2}{p_{N_2} \times p_{H_2}^3}$	1
1(d)(ii)	$K_p = 1.(00) \times 10^{-16}$	1
	Pa^{-2}	1
1(d)(iii)	(yield of ammonia) increases	1
	(value of K_p) stays the same	1

46. O/N 17/P22/Q2/b(ii),c,d

2(b)(ii)	(percentage decomposition of PCl_5) increases	1
	(forward) reaction is endothermic	1
2(c)	<u>rates</u> of forward and reverse / backward reactions are equal	1
	closed / sealed system/container	1
2(d)(i)	$n_{TOTAL} = 1.20 + 0.80 + 0.80$ OR 2.80 (mol) OR mole fraction = $1.20 / 2.80$ OR 0.429	1
	$p_{PCl_5} = 1 \times 10^5 \times (1.20 / 2.80) = 4.29 \times 10^4$ (Pa)	1
2(d)(ii)	$K_p = \frac{p_{PCl_3} \times p_{Cl_2}}{p_{PCl_5}}$	1
2(d)(iii)	1.91×10^4	1
	Pa	1

47. O/N 17/P23/Q1/b(iii),c,d

1(b)(iii)	reduces yield (of ammonia).	1
	(increasing T) shifts equilibrium (reaction) to the left / in the reverse direction / towards N_2 and H_2 / towards reactants / in endothermic direction	1
	to oppose the change OR oppose the increase in temperature OR to absorb the (additional) heat / energy OR decrease the temperature	1
1(c)(i)	$N_2 = 0.850$ (mol)	1
	$H_2 = 2.55$ (mol)	1
1(c)(ii)	$n_{TOTAL} = 3.7$ mol	1
	mol fraction of $NH_3 = 0.3 / 3.7$	1
	$p_{NH_3} = 2 \times 10^7 \times (0.3 / 3.7) = 1.62 \times 10^6$	1
1(d)(i)	$K_p = \frac{p_{NH_3}^2}{p_{N_2} \times p_{H_2}^3}$	1

1(d)(ii)	$K_p = 1.00 \times 10^{-16}$	1
	Pa^{-2}	1
1(d)(iii)	(yield of ammonia) increases	1
	(value of K_p) stays the same	1

48. M/J 17/P22/Q3/b(iii),c,d

3(b)(iii)	increases (%) decomposition (of HBr)	1
	(increasing T) shifts equilibrium to the right / in the forward direction / endothermic direction / towards $\text{H}_2 + \text{Br}_2$	1
	to oppose the change or oppose the increase in temperature OR to absorb (additional) energy / heat OR to decrease the temperature	1
3(c)(i)	$\text{H}_2 = 0.015$ (mol)	1
	$\text{HCl} = 0.27$ (mol)	1
3(c)(ii)	$\text{HCl} = 9/10$ AND $x\text{H}_2 = 1/20$ AND $\text{Cl}_2 = 1/20$ OR $\text{HCl} = 0.9(0)$ AND $\text{H}_2 = 0.05$ AND $\text{Cl}_2 = 0.05$	1
3(d)(i)	$(K_p =) \frac{p\text{H}_2 \times p\text{Cl}_2}{p\text{HCl}^2}$	1
3(d)(ii)	equal number of moles (of gas) on either side (of equation) / (total) pressure cancels	1
3(d)(iii)	4.649×10^{-3}	1

49. M/J 17/P23/Q3/b(iii),c,d

3(b)(iii)	(rate) increases AND correct explanation in terms of 'more collisions'	1
	more successful collisions per unit time / higher chance of successful collisions per unit time / higher proportion of successful collisions per unit time	1
	(yield) increases and shifts equilibrium to the right / in the forward direction / towards SO_3 / towards the product / in exothermic direction	1
	to oppose the change or oppose the increase in pressure / fewer molecules on RHS so eqm moves to right (to oppose change)	1
3(c)(i)	$\text{SO}_2 = 0.01$ (mol) AND $\text{SO}_3 = 0.99$ (mol)	1
3(c)(ii)	$n_{\text{TOT}} = 1.505$	1
	$p\text{O}_2 = 1.50 \times 10^5 \times (0.505 / 1.505) = 5.03 \times 10^4$ (Pa)	1
3(d)(i)	$(K_p =) \frac{p\text{SO}_3^2}{p\text{O}_2 \times p\text{SO}_2^2}$	1
3(d)(ii)	0.1946737305	1
	Pa^{-1}	1

50. MAR 17/P22/Q2/b(i),c(iii,iv,v,vi)

2(b)(i)	M1 base is Cl^- AND conjugate acid is HCl OR base is HSO_4^- AND conjugate acid is H_2SO_4					1																									
	M2 $\text{Cl}^-/\text{HSO}_4^-$ /base is a proton acceptor OR $\text{HCl}/\text{H}_2\text{SO}_4$ /(conjugate) acid has one more H^+					1																									
2(c)(iii)	reaction is exothermic					1																									
	(increased temperature) shifts equilibrium to the left AND decreases yield of products (Cl_2 and/or H_2O)/less product formed					1																									
2(c)(iv)	<table><tr><td></td><td>HCl</td><td>O_2</td><td>Cl_2</td><td>H_2O</td></tr><tr><td>initial number of moles</td><td>1.60</td><td>0.500</td><td>0</td><td>0</td></tr><tr><td>M1 eqm number of moles</td><td>$1.60 - 2 \times 0.600 = 0.400$</td><td>$0.500 - \frac{1}{2} \times 0.600 = 0.200$</td><td>0.600</td><td>0.600</td></tr><tr><td>M2 mole fraction</td><td></td><td></td><td>$\frac{0.600}{1.80}$</td><td></td></tr><tr><td>M3 partial pressure</td><td></td><td></td><td>$\frac{0.600}{1.80} \times p_{\text{tot}} = 5.00 \times 10^4$</td><td></td></tr></table>						HCl	O_2	Cl_2	H_2O	initial number of moles	1.60	0.500	0	0	M1 eqm number of moles	$1.60 - 2 \times 0.600 = 0.400$	$0.500 - \frac{1}{2} \times 0.600 = 0.200$	0.600	0.600	M2 mole fraction			$\frac{0.600}{1.80}$		M3 partial pressure			$\frac{0.600}{1.80} \times p_{\text{tot}} = 5.00 \times 10^4$		3
	HCl	O_2	Cl_2	H_2O																											
initial number of moles	1.60	0.500	0	0																											
M1 eqm number of moles	$1.60 - 2 \times 0.600 = 0.400$	$0.500 - \frac{1}{2} \times 0.600 = 0.200$	0.600	0.600																											
M2 mole fraction			$\frac{0.600}{1.80}$																												
M3 partial pressure			$\frac{0.600}{1.80} \times p_{\text{tot}} = 5.00 \times 10^4$																												
2(c)(v)	$K_p = \frac{(3.6 \times 10^4)^2 \times (3.6 \times 10^4)^2}{(4.8 \times 10^4)^4 \times 3.0 \times 10^4} = 1.05 \times 10^{-5}$					1																									
	units = Pa^{-1}					1																									
2(c)(vi)	K_p would not change					1																									

51. O/N 16/P22/Q2/b(ii,iii,iv,v,vi,vii)

2(b)(ii)	$2n$	1
2(b)(iii)	$0.05 - n + 2n$ OR $0.05 + n$	1
2(b)(iv)	$\frac{2n}{(0.05 + n)}$	1
2(b)(v)	$\text{N}_2\text{O}_4 = 0.0375$ $\text{NO}_2 = 0.0250$	1 1
2(b)(vi)	$K_p = \frac{p\text{NO}_2^2}{p\text{N}_2\text{O}_4}$	1

2(b)(vii)	$(0.4 \times 1.68 \times 10^5)^2 / (0.6 \times 1.68 \times 10^5)$ OR $0.4^2 \times 1.68 \times 10^5 / 0.6$	1
	44800 OR 44.8	1
	Pa OR kPa	1

52. M/J 16/P23/Q2/b(iv)

(iv)	production of ammonia is <u>exothermic</u> / (forward) reaction <u>exothermic</u> position of eqm would move to left / reverse / reduce yield (at higher T)	[1] [1]	[2]
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53. MAR 16/P22/Q3/b

(b) (i)	forward and backward reactions occurring <u>at same rate</u> OR <u>the rate of</u> forward and backward reactions are equal	[1]	[1]
(ii)	M1 = decreased yield of products / less products formed / ora M2 = <u>left</u> -hand side has fewer moles of gas OR equilibrium shifts to the <u>left</u>	[1] [1]	[2]

54. O/N 15/P21/Q2/b(ii,iii)

(ii)	Fe catalyst 200 atm 400–500 (°)C	[1] [1] [1]	[3]
(iii)	High T increases rate AND Low T improves yield owtte Chosen temp is a compromise High P favours / increases (both rate and) yield owtte pressure chosen limited by cost (of compression and 'thick walls')	[1] [1] [1] [1]	[4]

55. O/N 15/P23/Q2/b(ii,iii)

(ii)	Fe catalyst 200 atm 400–500 (°)C	[1] [1] [1]	[3]
(iii)	High T increases rate AND Low T improves yield owtte Chosen temp is a compromise High P favours / increases (both rate and) yield owtte pressure chosen limited by cost (of compression and 'thick walls')	[1] [1] [1] [1]	[4]

56. M/J 15/P22/Q2/b

(b) (i)	T = 400 – 600 °C (chosen as a compromise because) High T increases rate ora High T decreases yield / moves eqm left / makes less SO ₃ as forward reaction exothermic ora	[1] [1] [1]	[3]
(ii)	High pressure increases rate as collision frequency increases ora High pressure moves eqm right / favours forward reaction as more moles on left ora Uneconomic to use high pressures / high yield at low pressure	[1] [1] [1]	[3]